

5G: Architecture and Key Principles

5G RAN - INTRODUCTION



5G RAN

- New Spectrum
- New use cases



5G RAN

- Enhanced MBB: Wideband carriers, carrier aggregation

- URLLC: Flexible numerology, Preemptive scheduling, grant free uplink etc

- Cost efficiency: Virtualization



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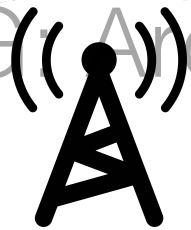
5G: Architecture and Key Principles

5G RAN – MASSIVE MIMO



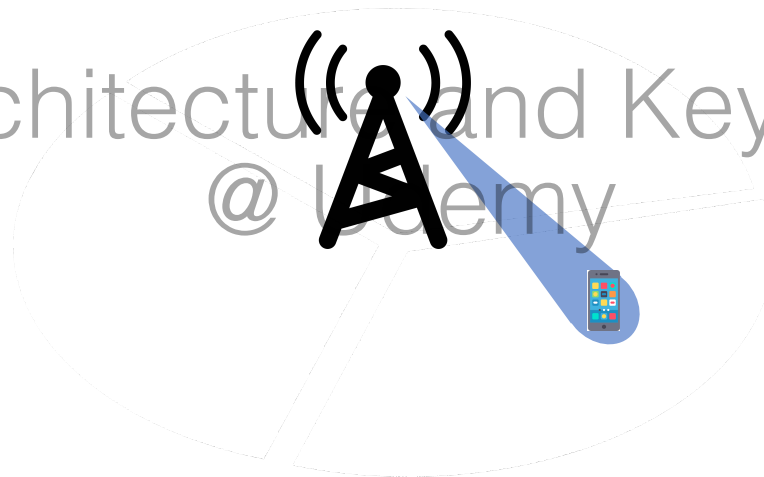
Background

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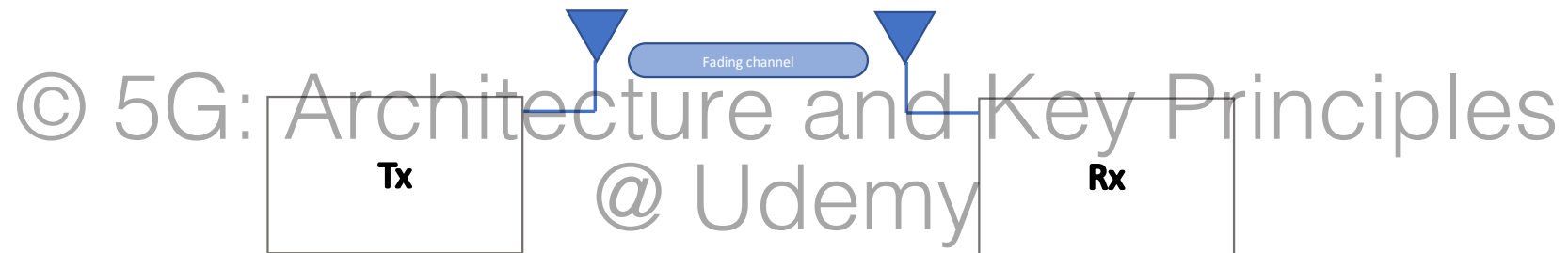


Background

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SISO

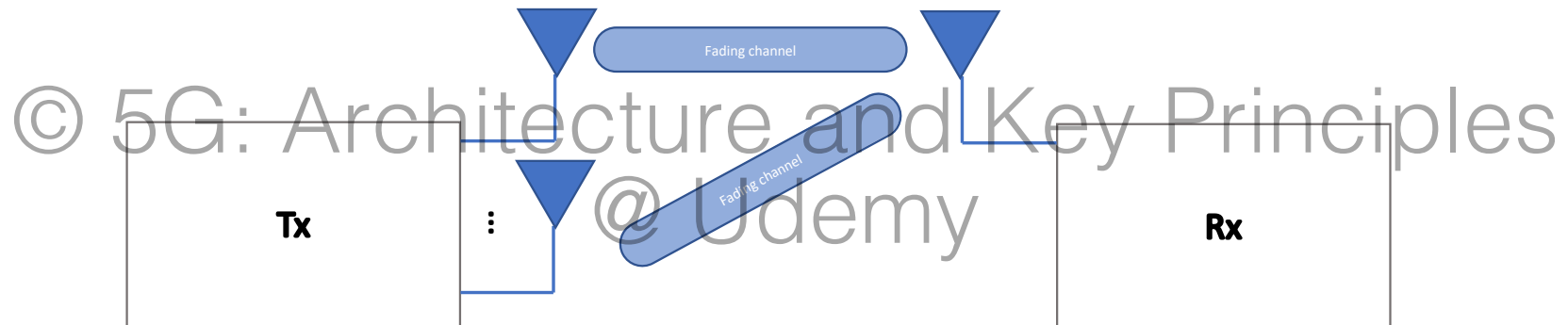


$$C = B \log_2 \left(1 + \frac{S}{N} \right)$$

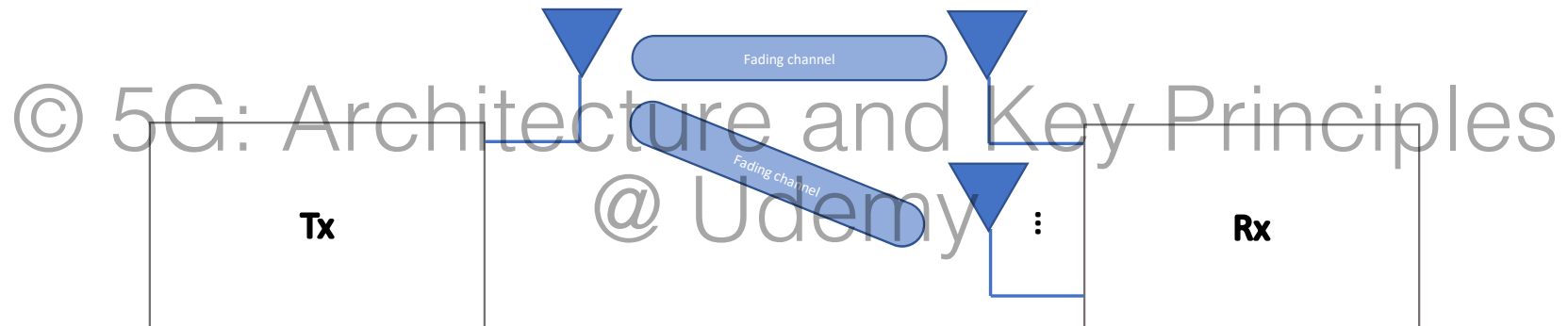
Figure 1: The Famous Shannon Limit/Capacity Equation



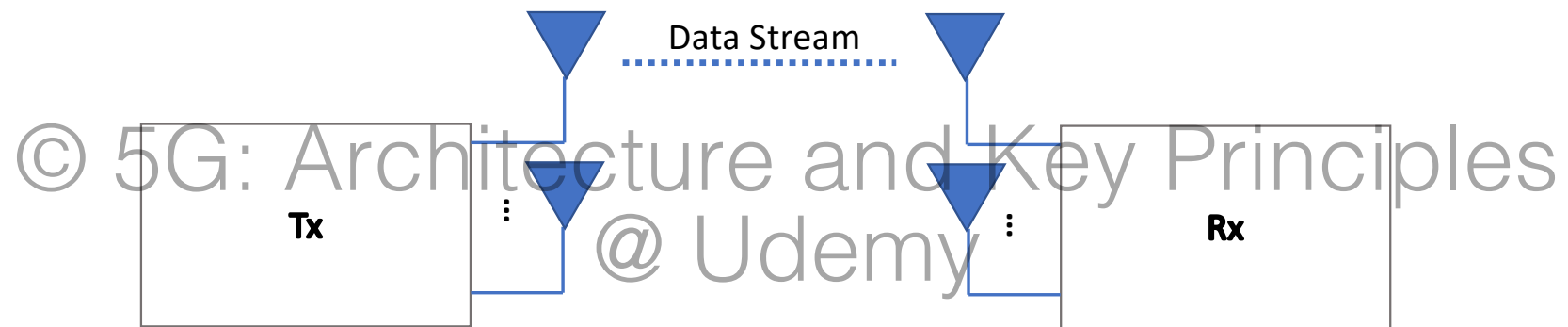
MISO – Transmit Diversity



MISO – Receive Diversity



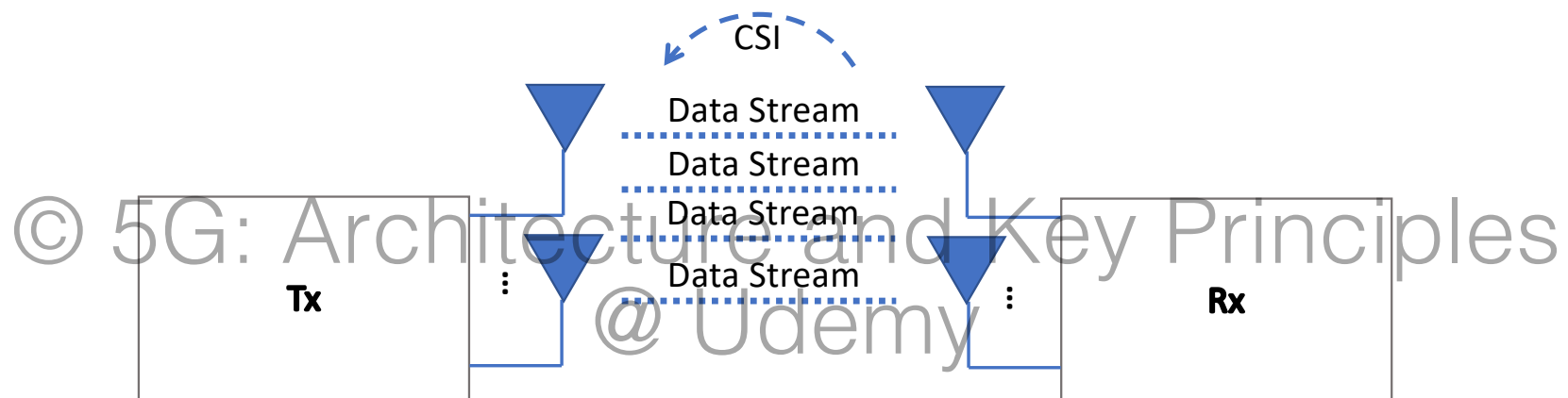
MIMO – Diversity techniques



Low SNR – Diversity techniques to improve SNR



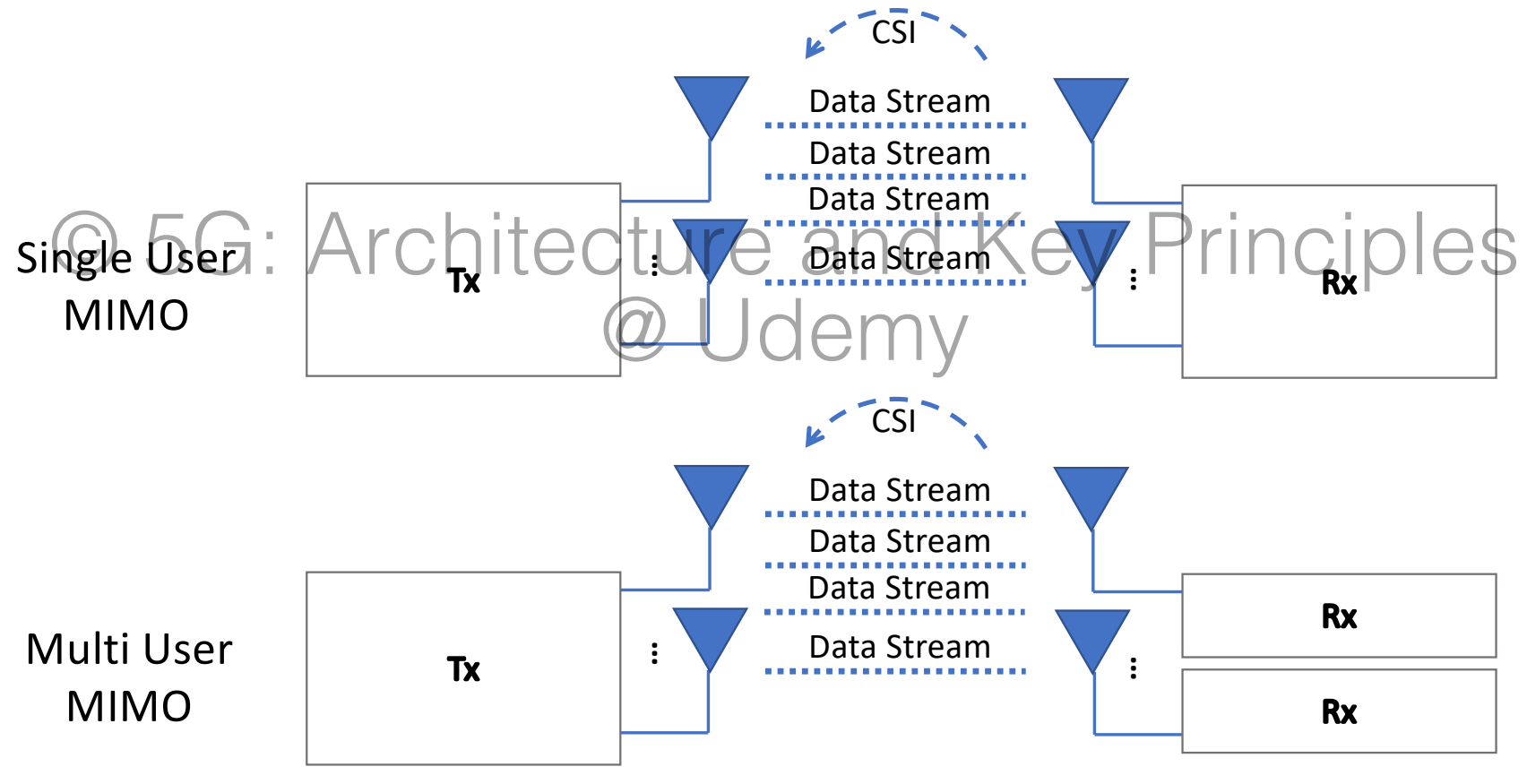
MIMO – Spatial Multiplexing



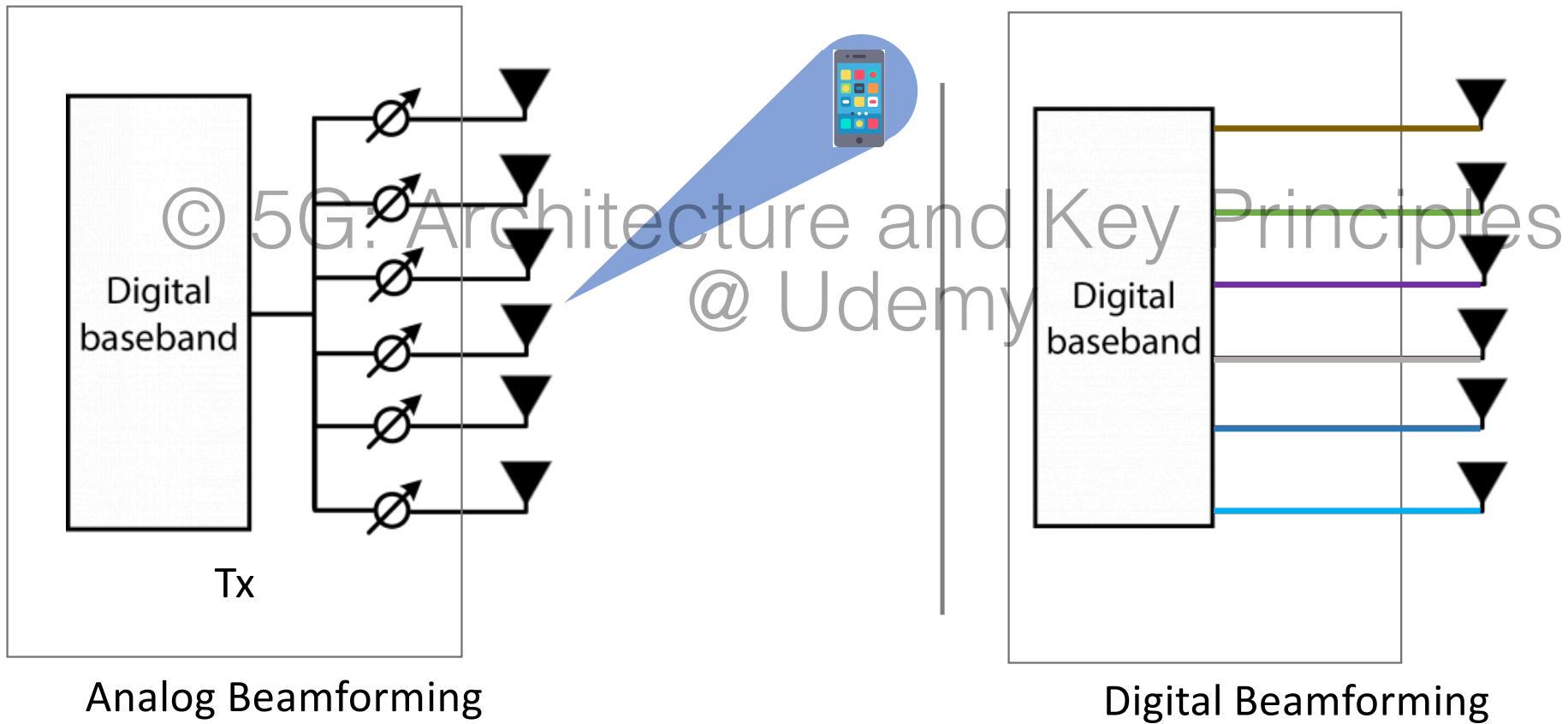
- High SNR – Spatial multiplexing to improve throughput
- Open Loop – Tx has no CSI (Channel State Information)
 - Closed Loop – Rx feeds back CSI
- Max. Streams = $\min(\text{\#TxAntennas}, \text{\#RxAntennas})$



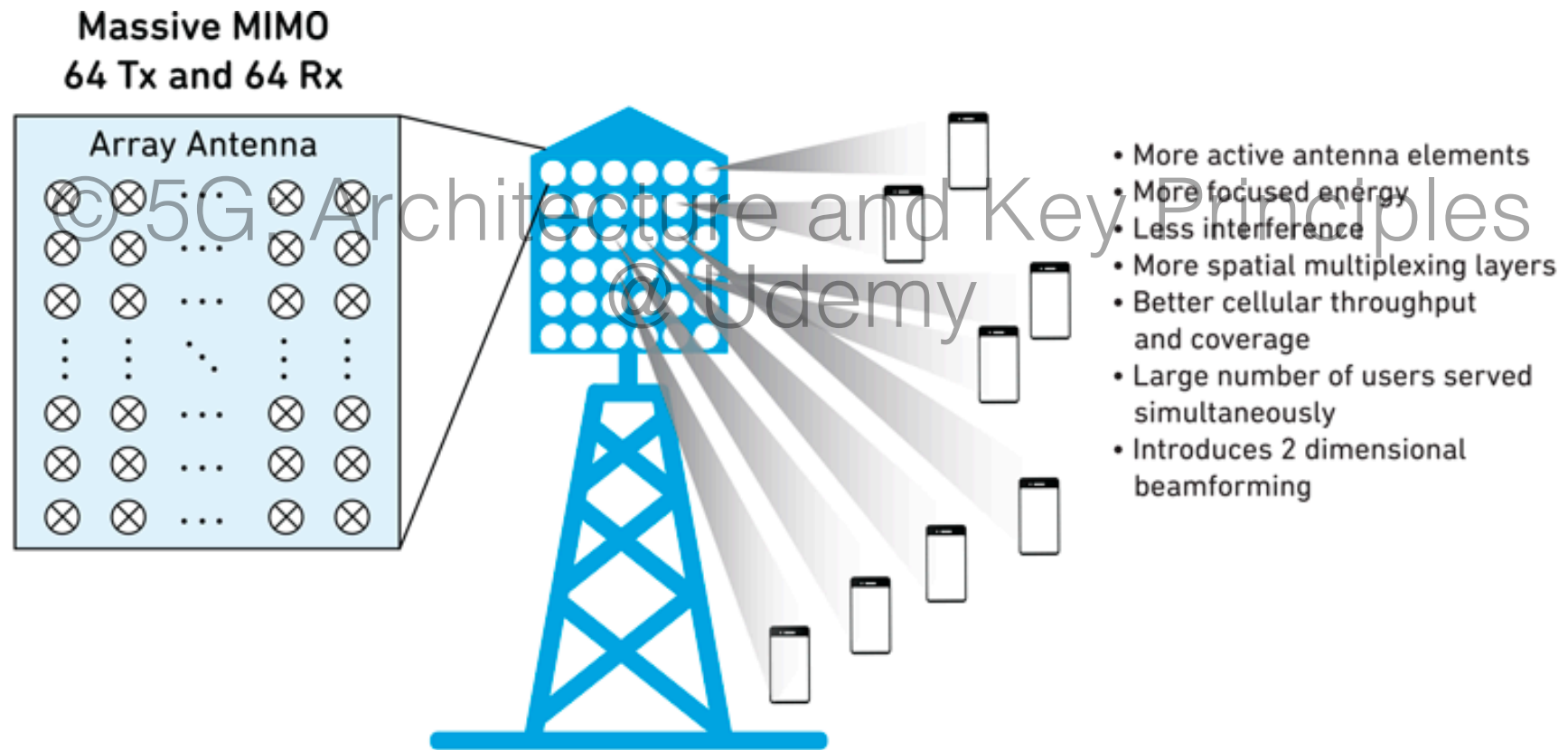
MIMO – Spatial Multiplexing



MIMO – Beamforming



Massive MIMO



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5G RAN – SPECTRUM



NR Spectrum

NR operating Band	Uplink (UL) operating band	Downlink (DL) operating band	Duplex Mode	$\Delta F_{\text{Raster}}(\text{kHz})$	Uplink	Downlink
	BS receive / UE transmit $F_{\text{UL,low}} - F_{\text{UL,high}}$	BS transmit / UE receive $F_{\text{DL,low}} - F_{\text{DL,high}}$			Range of N_{REF} (First – <Step size> – Last)	Range of N_{REF} (First – <Step size> – Last)
n1	1920 MHz – 1980 MHz	2110 MHz – 2170 MHz	FDD	100	384000 – <20> – 396000	422000 – <20> – 434000
n2	1850 MHz – 1910 MHz	1930 MHz – 1990 MHz	FDD	100	370000 – <20> – 382000	386000 – <20> – 398000
n3	1710 MHz – 1785 MHz	1805 MHz – 1880 MHz	FDD	100	342000 – <20> – 357000	361000 – <20> – 376000
n5	824 MHz – 849 MHz	869 MHz – 894 MHz	FDD	100	164800 – <20> – 169800	173800 – <20> – 178800
n7	2500 MHz – 2570 MHz	2620 MHz – 2690 MHz	FDD	100	500000 – <20> – 514000	524000 – <20> – 538000
n8	880 MHz – 915 MHz	925 MHz – 960 MHz	FDD	100	176000 – <20> – 183000	185000 – <20> – 192000
n12	699 MHz – 716 MHz	729 MHz – 746 MHz	FDD	100	139800 – <20> – 143200	145800 – <20> – 149200
n20	832 MHz – 862 MHz	791 MHz – 821 MHz	FDD	100	166400 – <20> – 172400	158200 – <20> – 164200
n25	1850 MHz – 1915 MHz	1930 MHz – 1995 MHz	FDD	100	370000 – <20> – 383000	386000 – <20> – 399000
n28	703 MHz – 748 MHz	758 MHz – 803 MHz	FDD	100	140800 – <20> – 149600	151600 – <20> – 160600
n34	2010 MHz – 2025 MHz	2010 MHz – 2025 MHz	TDD	100	402000 – <20> – 406000	402000 – <20> – 406000
n38	2570 MHz – 2620 MHz	2570 MHz – 2620 MHz	TDD	100	514000 – <20> – 524000	514000 – <20> – 524000
n39	1880 MHz – 1920 MHz	1880 MHz – 1920 MHz	TDD	100	376000 – <20> – 384000	376000 – <20> – 384000
n40	2300 MHz – 2400 MHz	2300 MHz – 2400 MHz	TDD	100	460000 – <20> – 480000	460000 – <20> – 480000
n41	2496 MHz – 2690 MHz	2496 MHz – 2690 MHz	TDD	15	499200 – <3> – 537999	499200 – <3> – 537999
				30	499200 – <6> – 537996	499200 – <6> – 537996
n50	1432 MHz – 1517 MHz	1432 MHz – 1517 MHz	TDD	100	286400 – <20> – 303400	286400 – <20> – 303400
n51	1427 MHz – 1432 MHz	1427 MHz – 1432 MHz	TDD	100	285400 – <20> – 286400	285400 – <20> – 286400
n65	1920 MHz – 2010 MHz	2110 MHz – 2200 MHz	FDD	100	384000 – <20> – 402000	422000 – <20> – 440000
n66	1710 MHz – 1780 MHz	2110 MHz – 2200 MHz	FDD	100	342000 – <20> – 356000	422000 – <20> – 440000
n70	1695 MHz – 1710 MHz	1995 MHz – 2020 MHz	FDD	100	339000 – <20> – 342000	399000 – <20> – 404000
n71	663 MHz – 698 MHz	617 MHz – 652 MHz	FDD	100	132600 – <20> – 139600	123400 – <20> – 130400
n74	1427 MHz – 1470 MHz	1475 MHz – 1518 MHz	FDD	100	285400 – <20> – 294000	295000 – <20> – 303600
n75	N/A	1432 MHz – 1517 MHz	SDL	100	N/A	286400 – <20> – 303400
n76	N/A	1427 MHz – 1432 MHz	SDL	100	N/A	285400 – <20> – 286400
n77	3300 MHz – 4200 MHz	3300 MHz – 4200 MHz	TDD	15	620000 – <1> – 680000	620000 – <1> – 680000
				30	620000 – <2> – 680000	620000 – <2> – 680000
n78	3300 MHz – 3800 MHz	3300 MHz – 3800 MHz	TDD	15	620000 – <1> – 653333	620000 – <1> – 653333
				30	620000 – <2> – 653332	620000 – <2> – 653332
n79	4400 MHz – 5000 MHz	4400 MHz – 5000 MHz	TDD	15	693334 – <1> – 733333	693334 – <1> – 733333
				30	693334 – <2> – 733332	693334 – <2> – 733332
n80	1710 MHz – 1785 MHz	N/A	SUL	100	342000 – <20> – 357000	N/A
n81	880 MHz – 915 MHz	N/A	SUL	100	176000 – <20> – 183000	N/A
n82	832 MHz – 862 MHz	N/A	SUL	100	166400 – <20> – 172400	N/A
n83	703 MHz – 748 MHz	N/A	SUL	100	140600 – <20> – 149600	N/A
n84	1920 MHz – 1980 MHz	N/A	SUL	100	384000 – <20> – 396000	N/A
n86	1710 MHz – 1780 MHz	N/A	SUL	100	342000 – <20> – 356000	N/A

Frequency Range 1
(Below 6 GHz)

NR operating band	Uplink (UL) and Downlink (DL) operating band	Duplex Mode	Uplink and Downlink	
	BS transmit/receive UE transmit/receive $F_{\text{UL,low}} - F_{\text{UL,high}}$ $F_{\text{DL,low}} - F_{\text{DL,high}}$		ΔF_{Raster}	Range of N_{REF} (First – <Step size> – Last)
n257	26500 MHz – 29500 MHz	TDD	60	2054166 – <1> – 2104165
			120	2054167 – <2> – 2104165
n258	24250 MHz – 27500 MHz	TDD	60	2016667 – <1> – 2070832
			120	2016667 – <2> – 2070831
n260	37000 MHz – 40000 MHz	TDD	60	2229166 – <1> – 2279165
			120	2229167 – <2> – 2279165
n261	27500 MHz – 28350 MHz	TDD	60	2070833 – <1> – 2084999
			120	2070833 – <2> – 2084999

Frequency Range 2
(Above 24 GHz)



Spectrum evolution

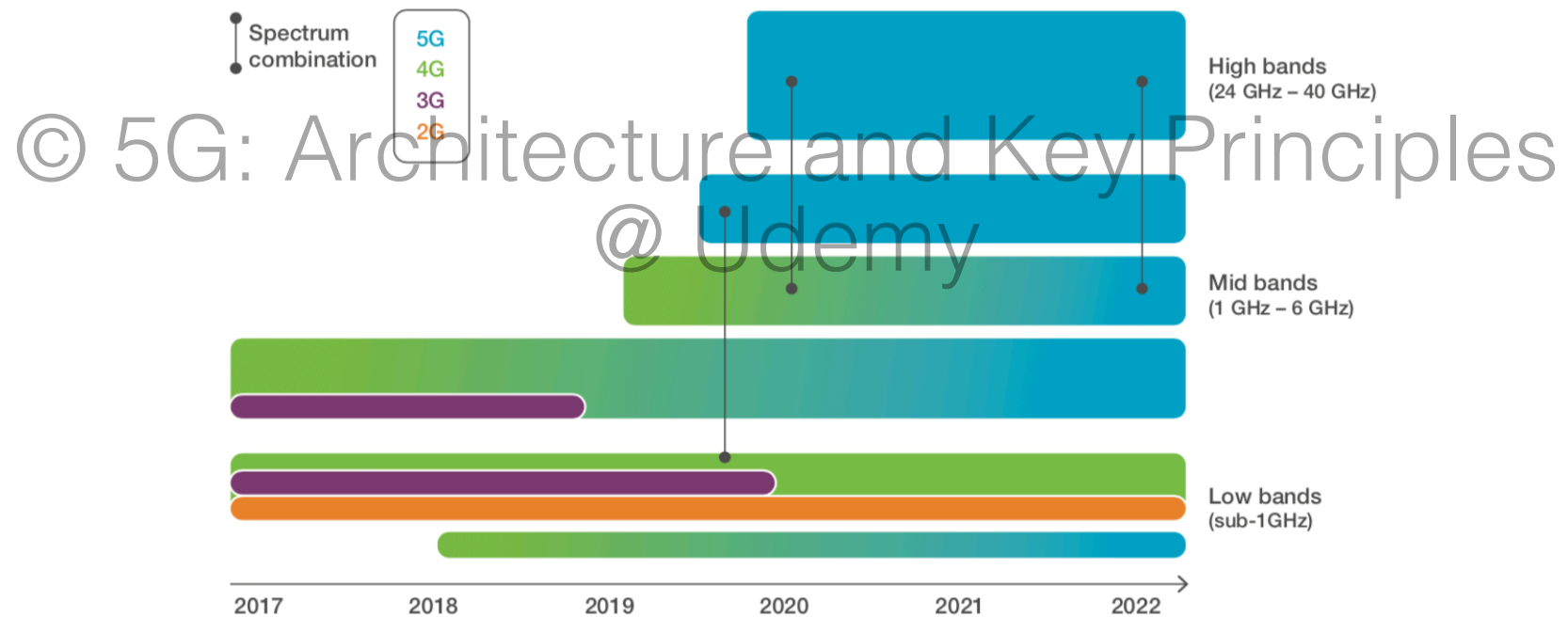


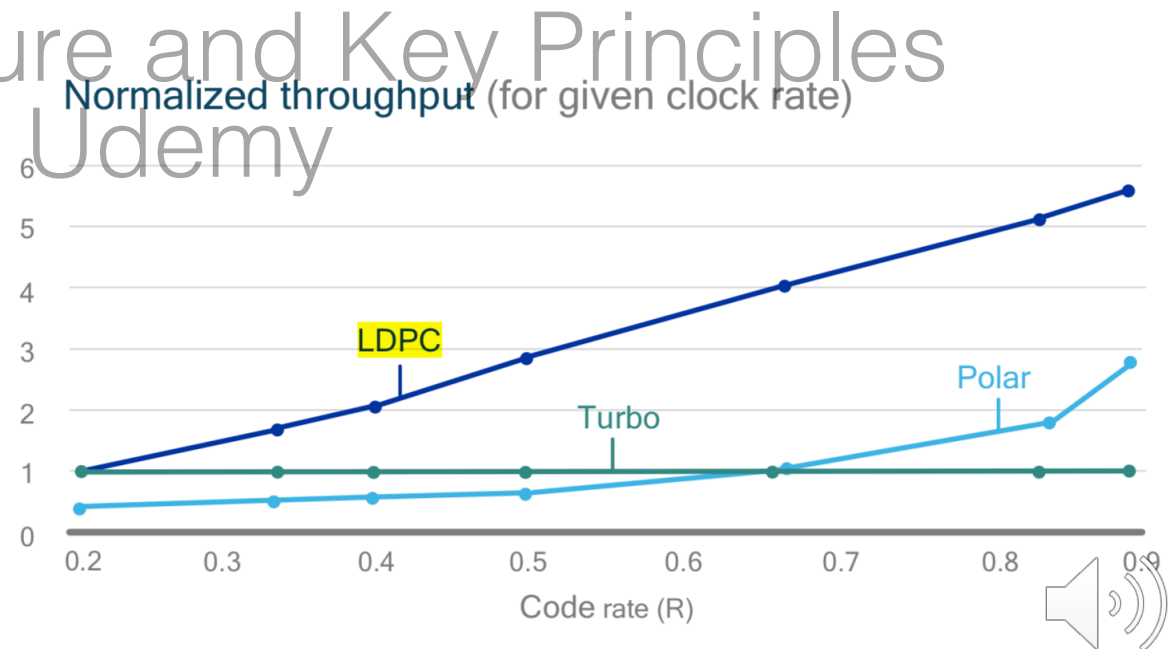
Figure 3: Spectrum allocation over time.



Spectral efficiency

- Example 1:
 - LTE: 20 MHz => 100 subcarriers of 15 kHz
 - NR: 20 MHz => 106 subcarriers of 15 kHz – Smaller guardbands

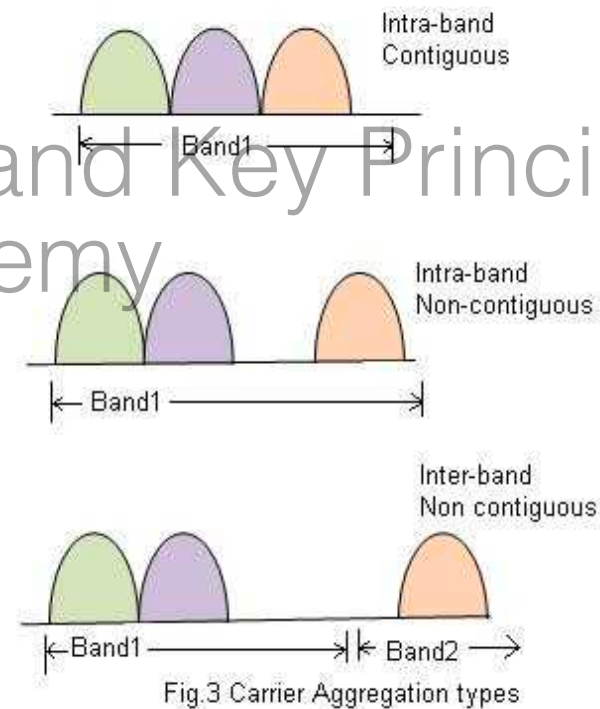
- Example 2:
 - LTE: Turbocodes
 - NR:
 - Datachannels: LDPC codes
 - Control channels: Polar codes



Carrier Aggregation (CA)

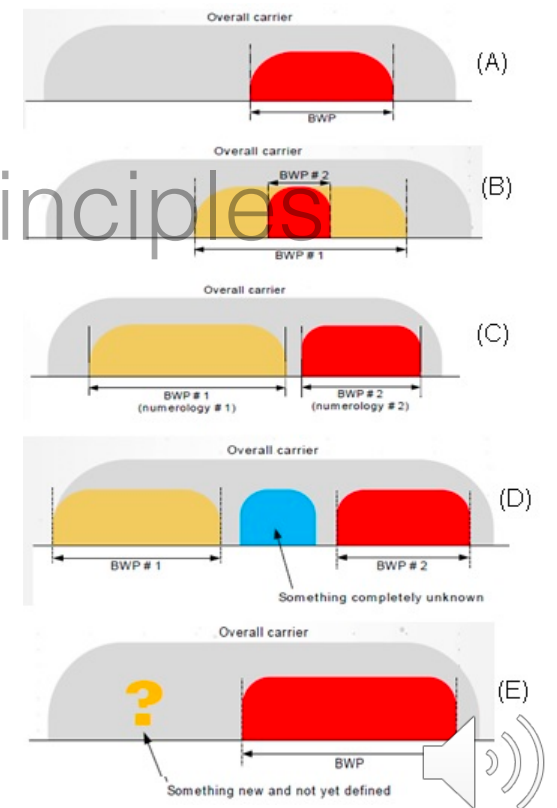
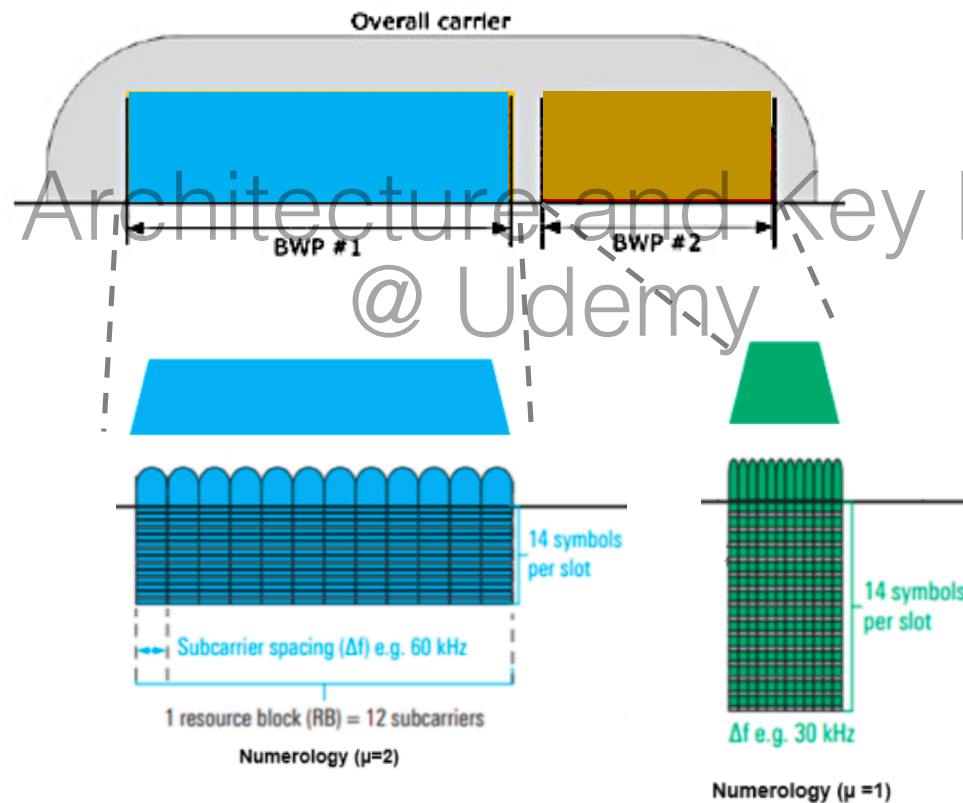
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- Upto 16 carriers
- Up to 1 GHz



Bandwidth parts (BWP)

Bandwidth parts: a subset of contiguous RBs on a carrier



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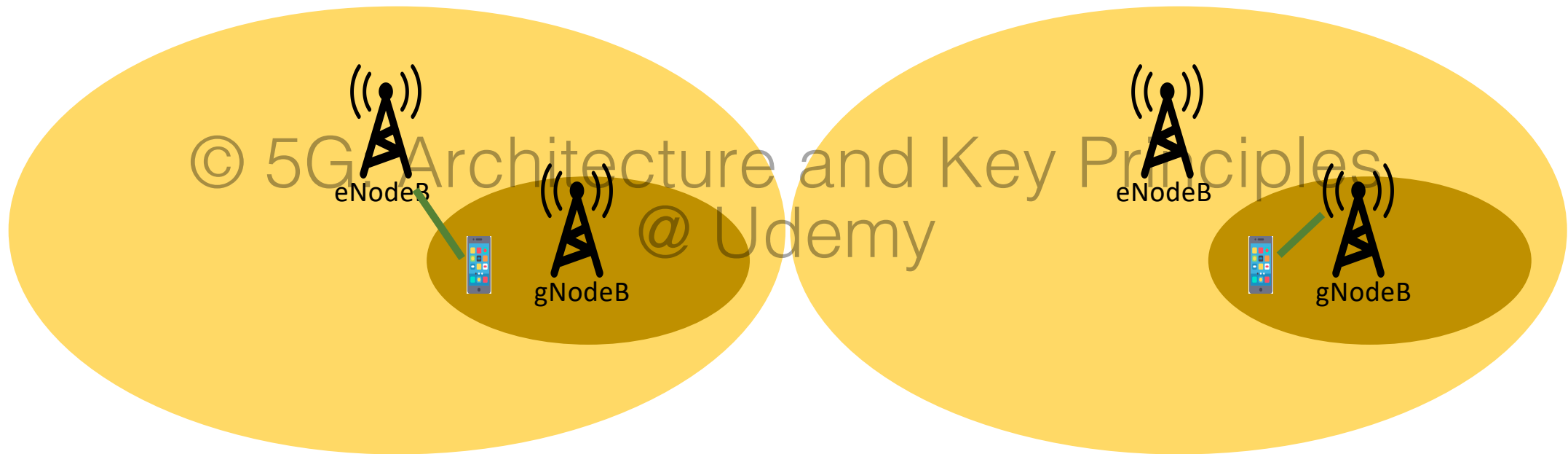


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5G RAN – DUAL
CONNECTIVITY



Dual Connectivity

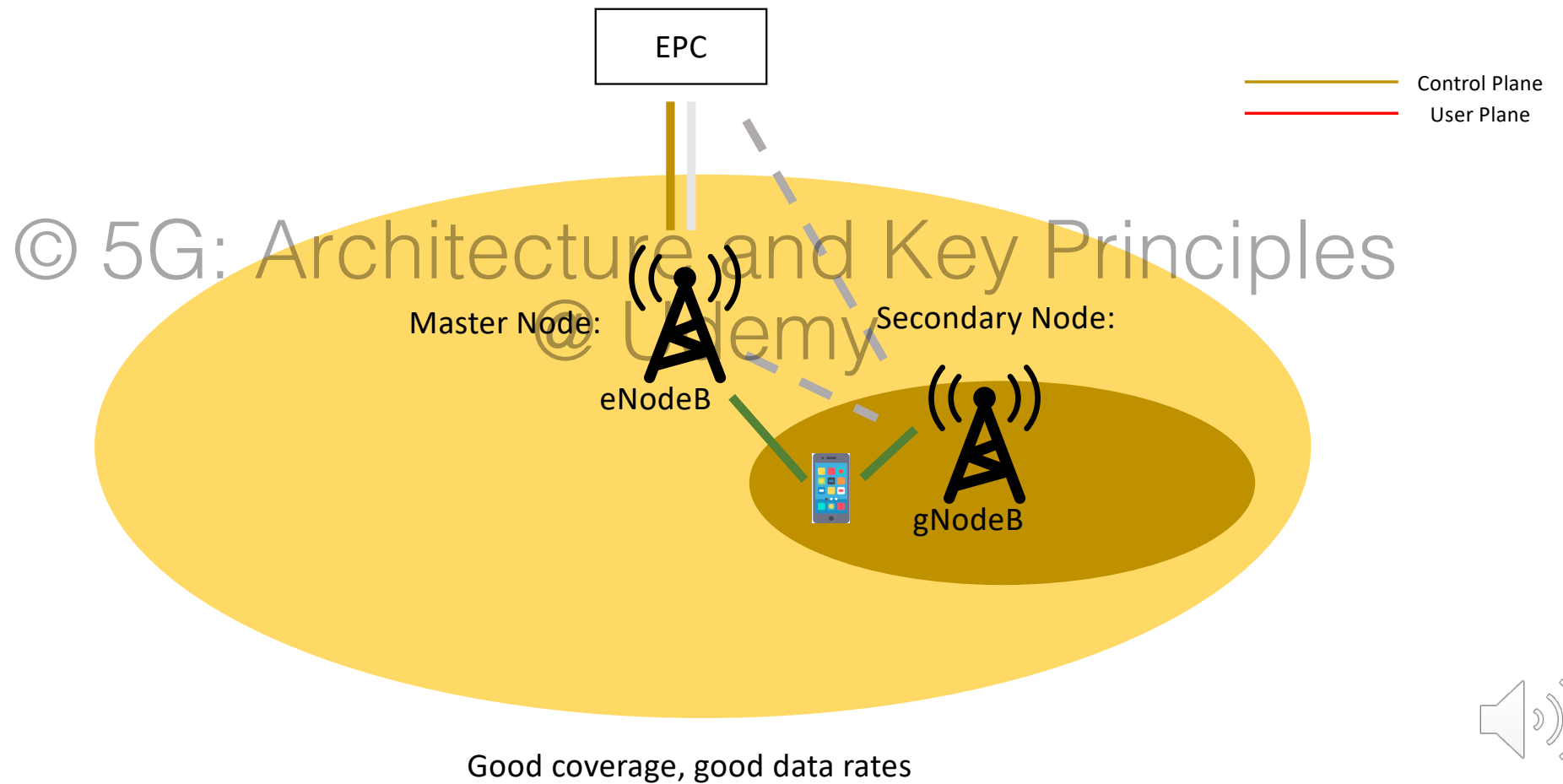


Good coverage, lower data rates

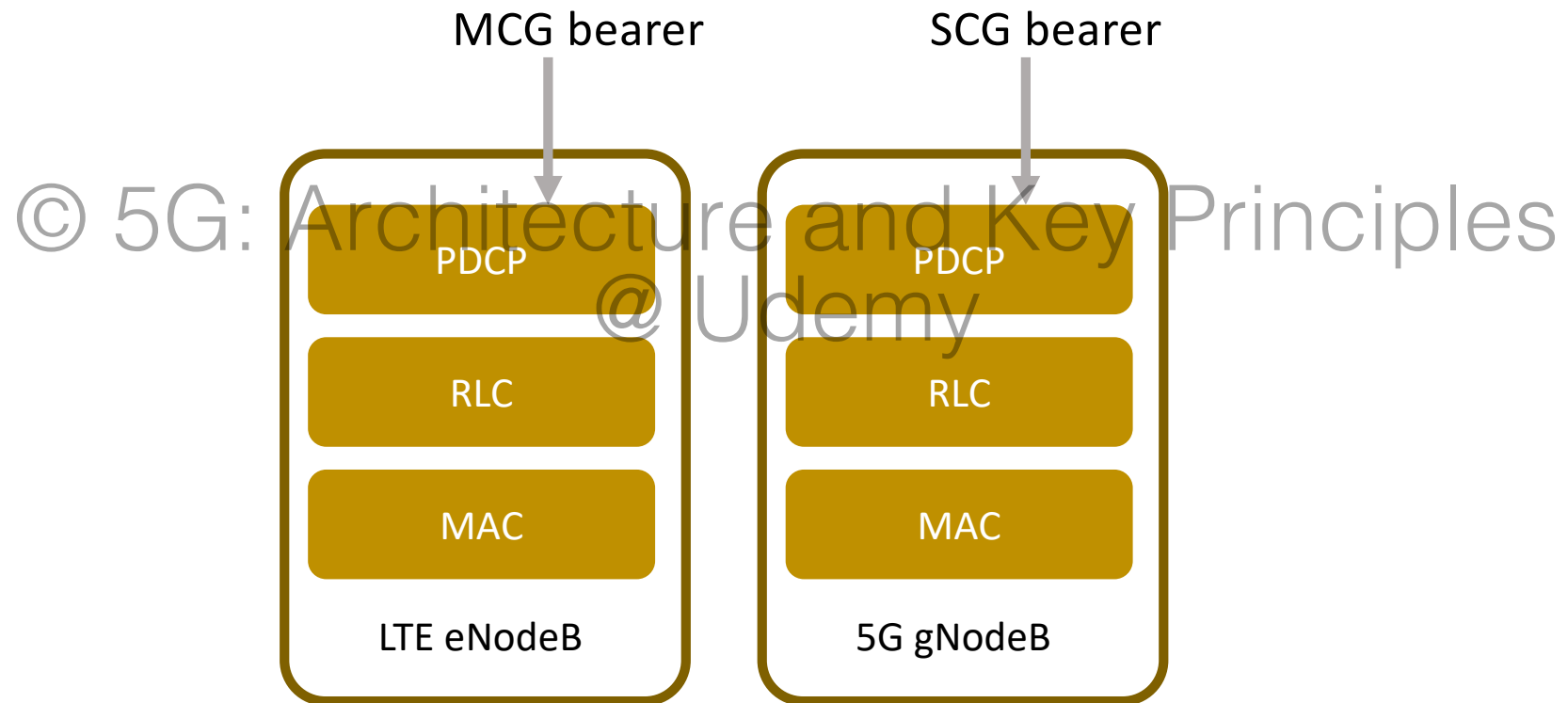
Worse coverage, better data rates



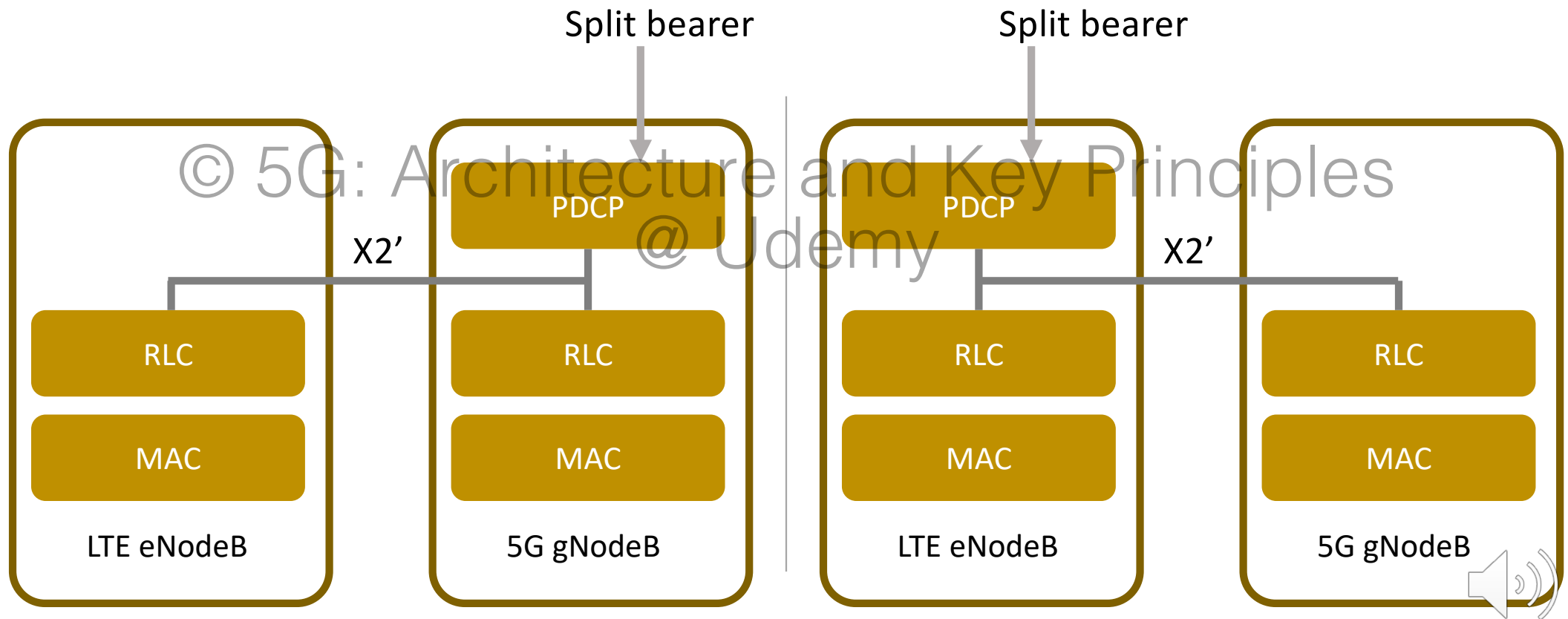
Dual Connectivity



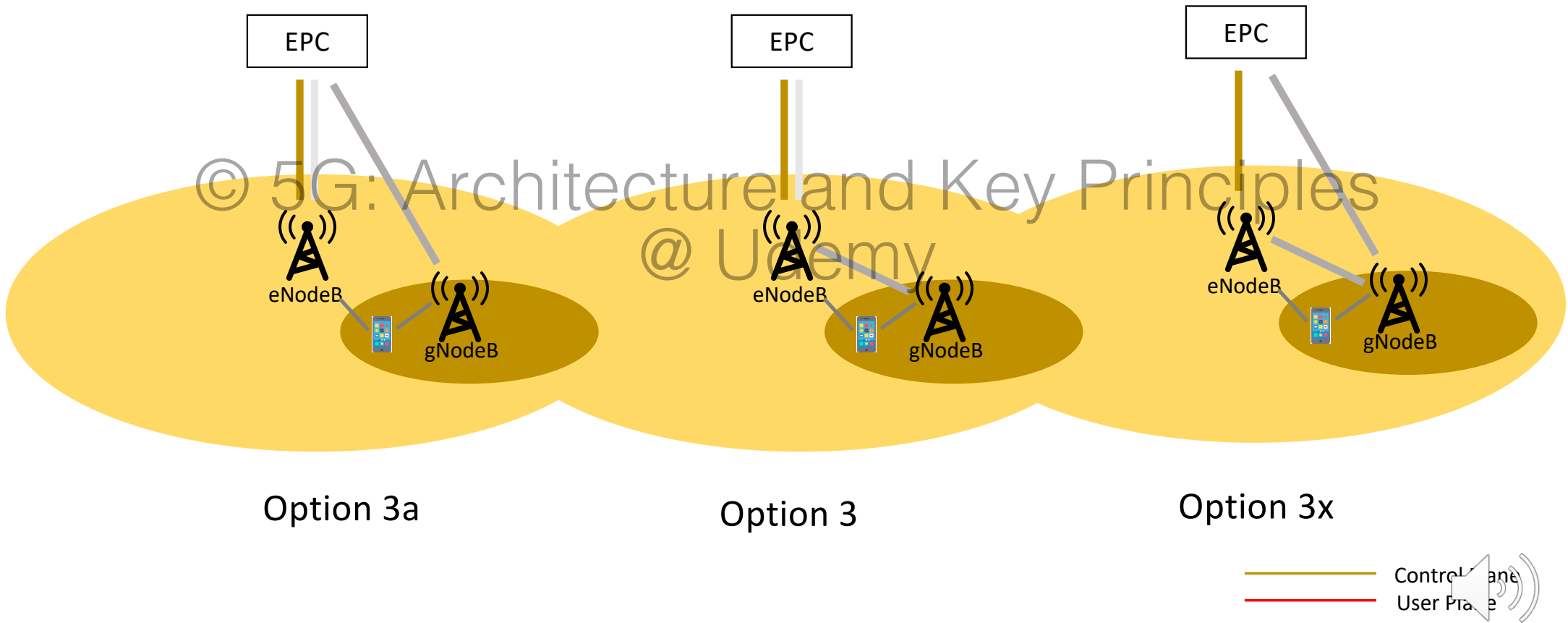
Dual Connectivity



Dual Connectivity



Dual Connectivity



Dual Connectivity

- Fast Load balancing
- Improved mobility robustness
- Higher per user throughput



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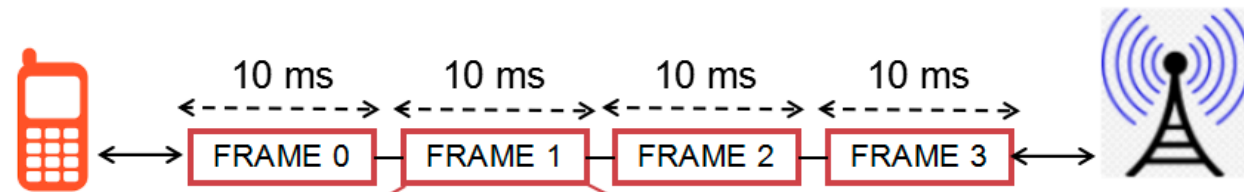


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5G RAN – FLEXIBLE
NUMEROLOGY

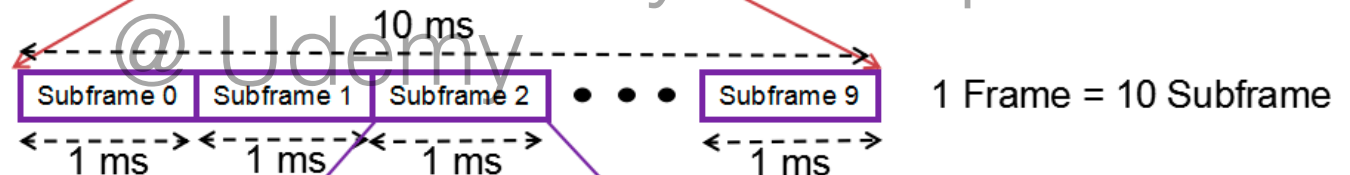


LTE Frame

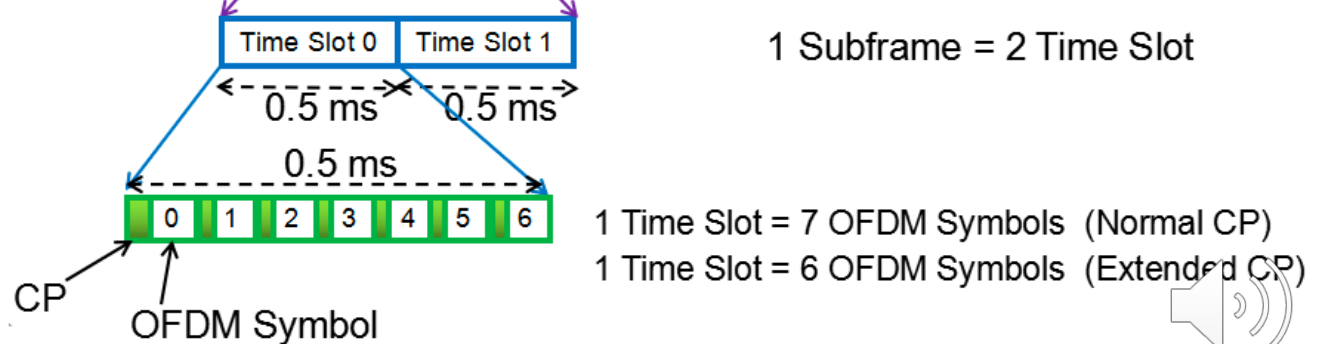


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LTE Frame structure:



Symbol duration: 66us

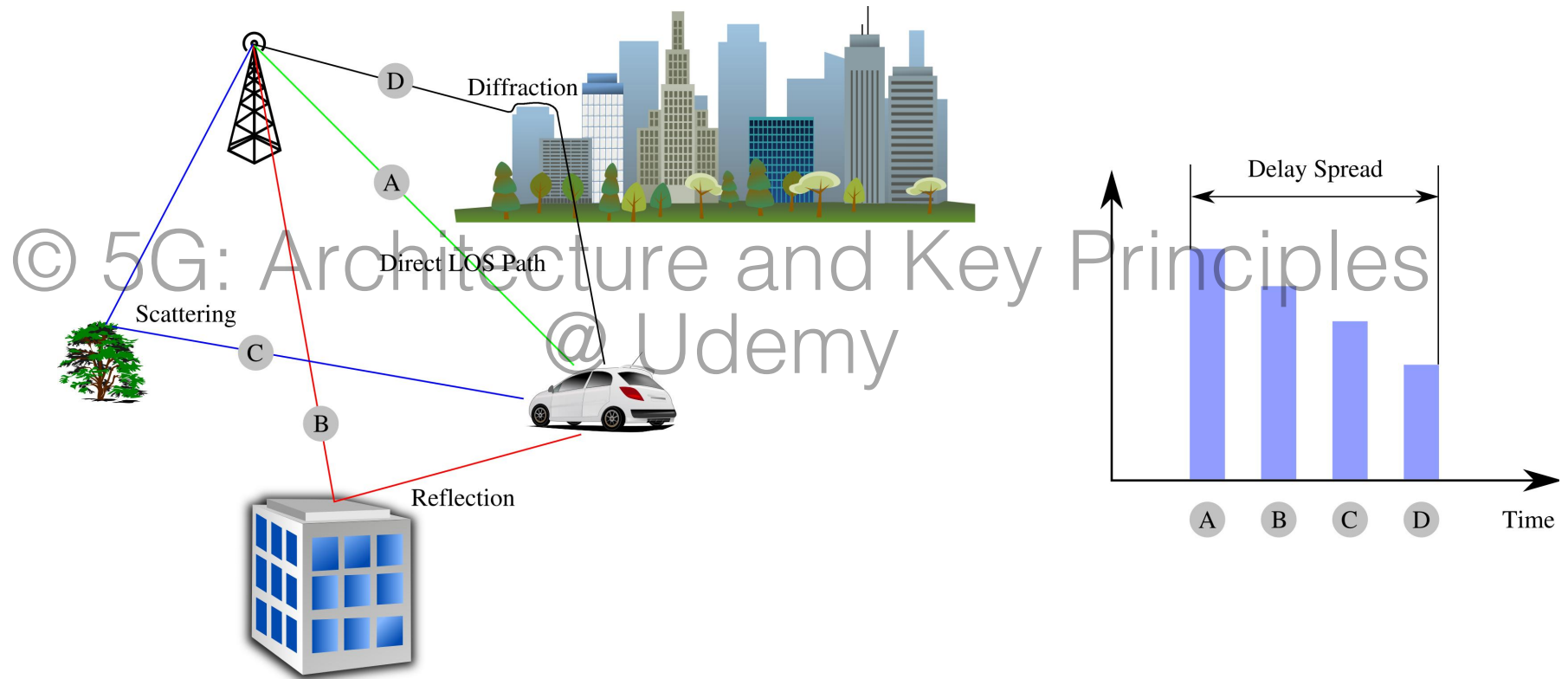


How to choose symbol duration?

Delay Spread and Coherence time



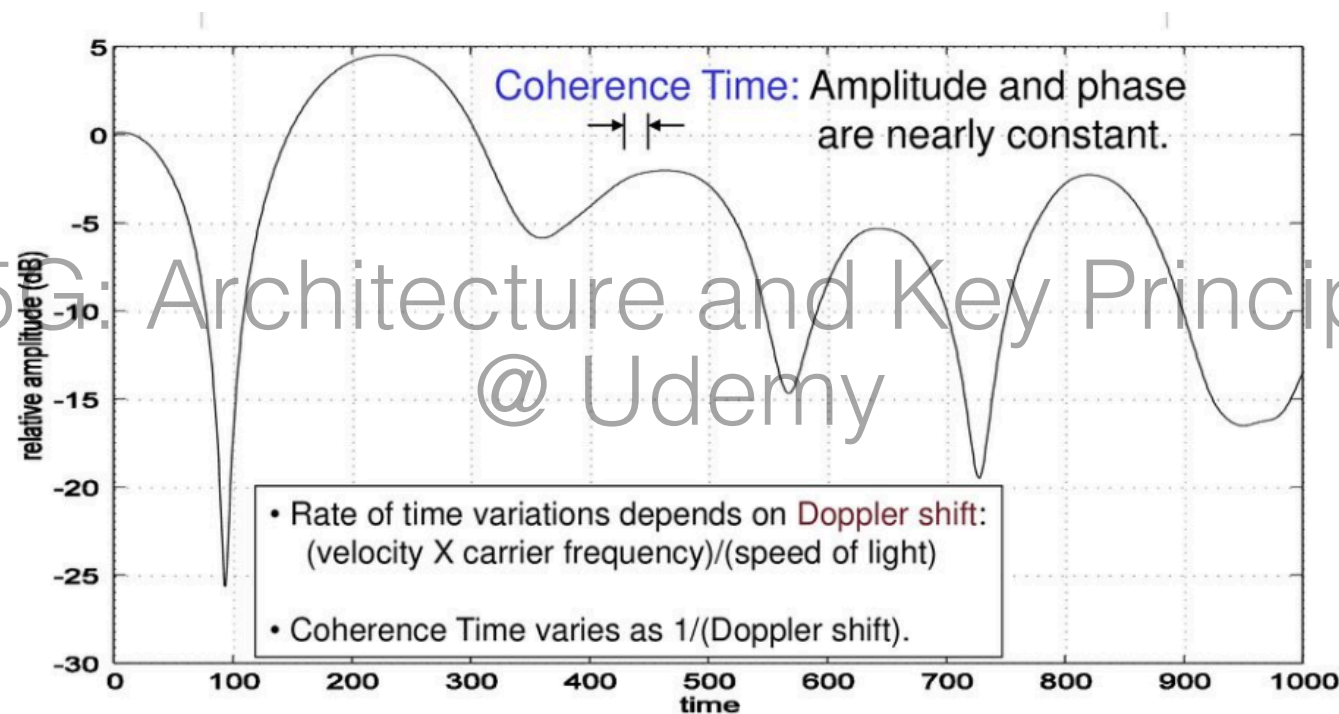
Delay Spread



- Total time interval during which reflections with significant energy arrive
- Larger cells => Larger delay spread



Coherence time



- Time duration over which the channel's response is considered to be not varying
- Decreases with higher frequency bands and faster moving UEs



Symbol duration

Delay spread $\ll$ Symbol duration $\ll$ Coherence time

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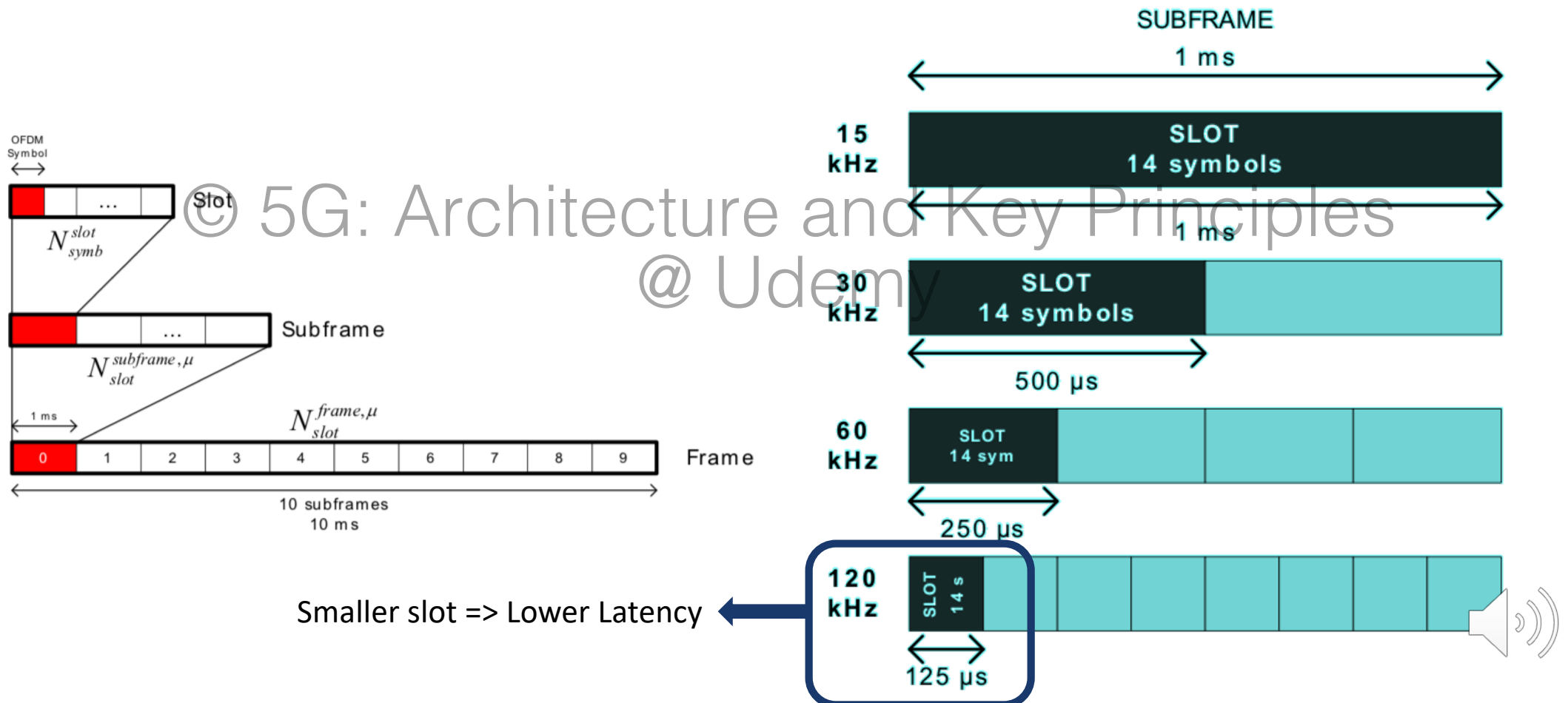
Varying cell sizes

Higher bandwidths
Fast moving UEs

Flexible Numerology



Flexible Numerology



Flexible Numerology

Outdoor and
macro coverage

FDD/TDD <3 GHz



Outdoor and
small cell

TDD > 3 GHz



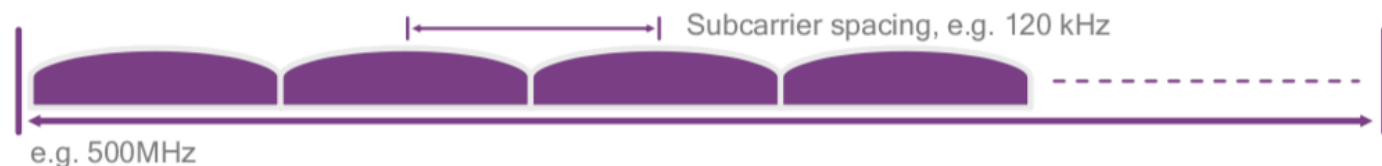
Indoor
wideband

TDD e.g. 5 GHz (Unlicensed)



mmWave

TDD e.g. 28 GHz



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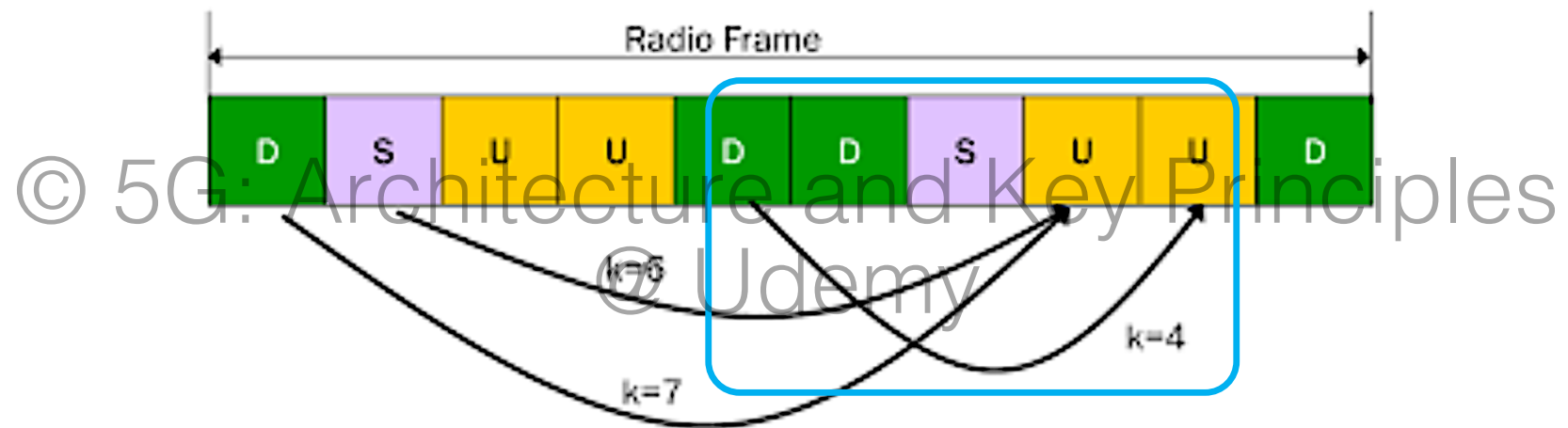


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5G RAN – MINI SLOT AND
PREEMPTIVE SCHEDULING



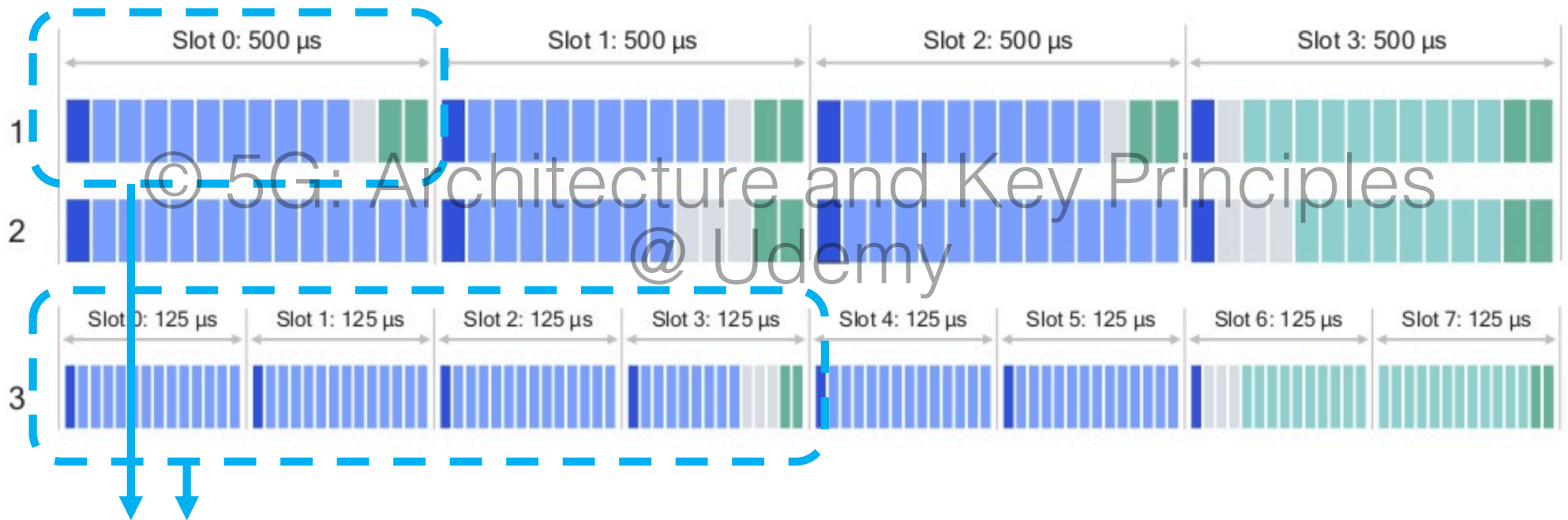
LTE Data to ACK/NACK



Minimum of 4 ms



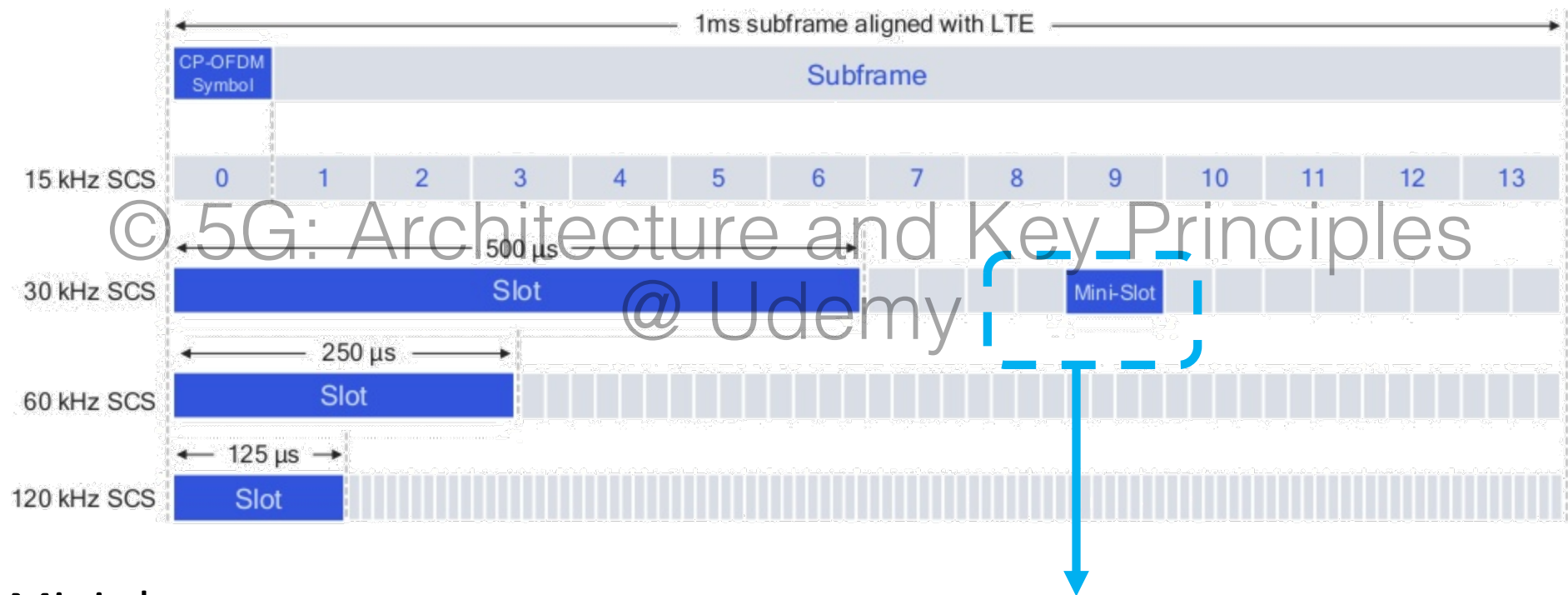
Self contained slot



Data and the corresponding ACK/NACK can all be fit in 500us in different numerology



Mini Slot

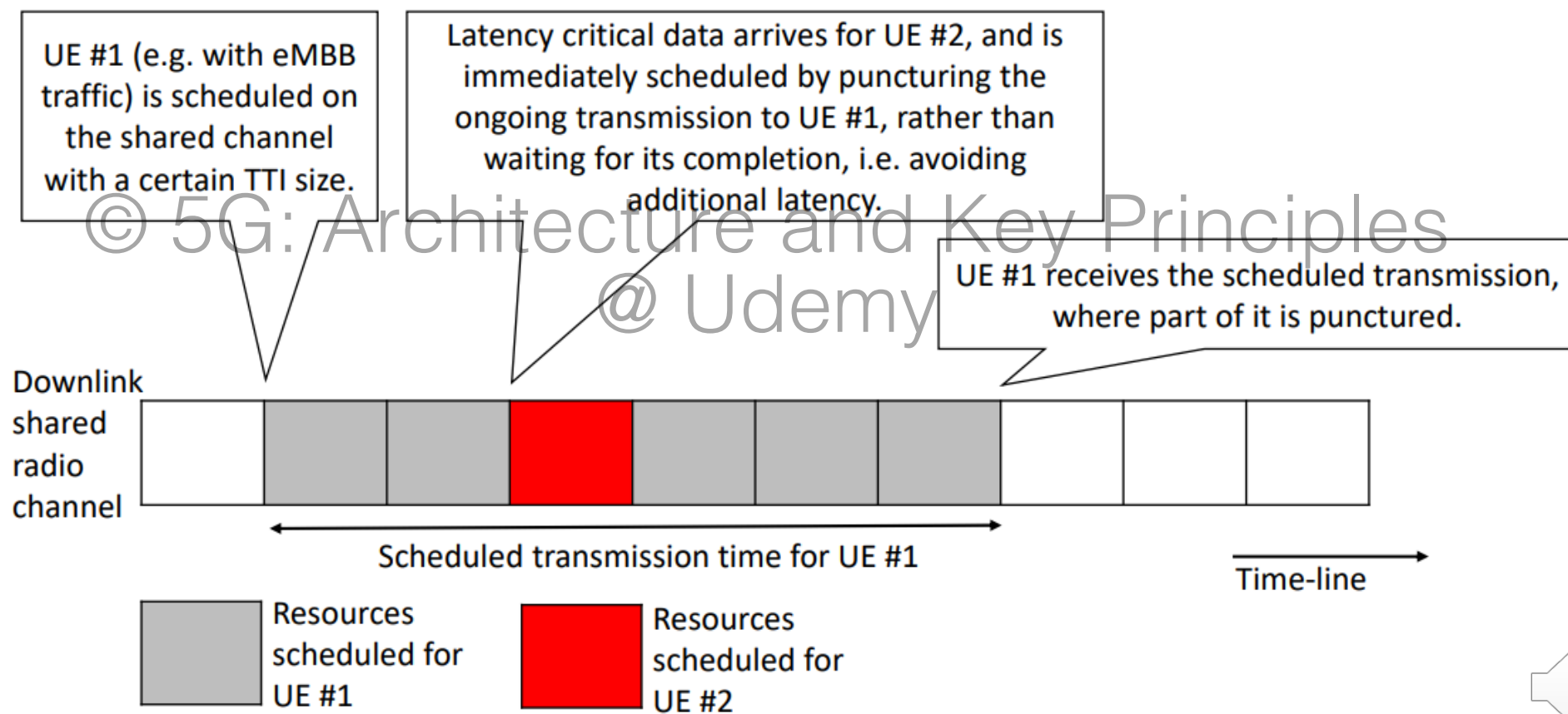


Mini slots:

- 2, 4 or 7 symbols long
- Can begin in symbol boundary instead of slot boundary



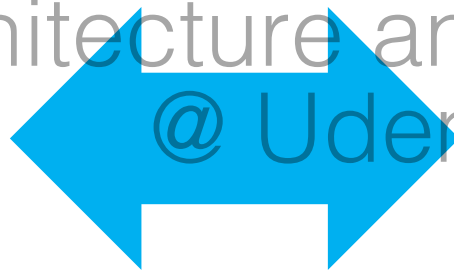
Preemptive Scheduling



Mini Slot tradeoff

Mini slot:

- Low latency
- But
 - Constant DCI monitoring
 - Higher energy consumption



No Mini slot:

- Normal latency
- But
 - Microsleep between DCI monitoring
 - Lower energy consumption



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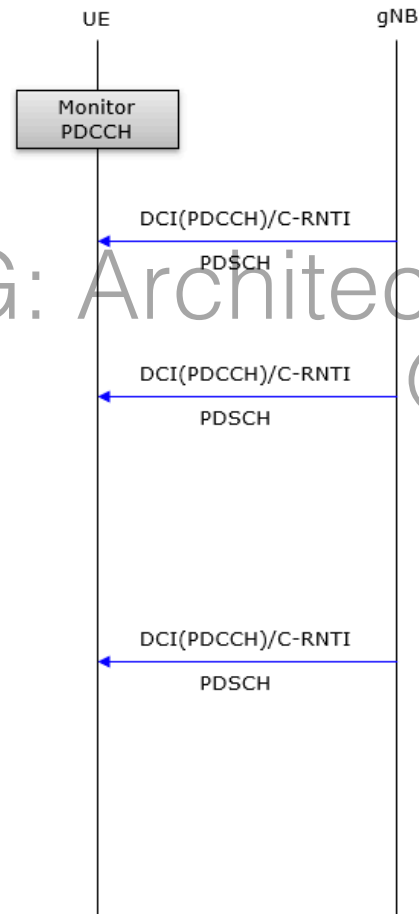
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5G RAN — SCHEDULING



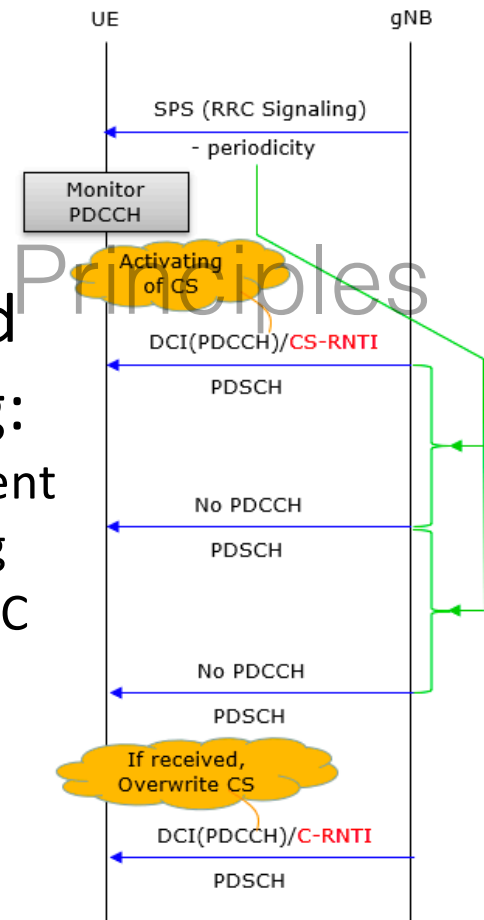
Downlink Scheduling

Dynamic Scheduling: - Through DCI

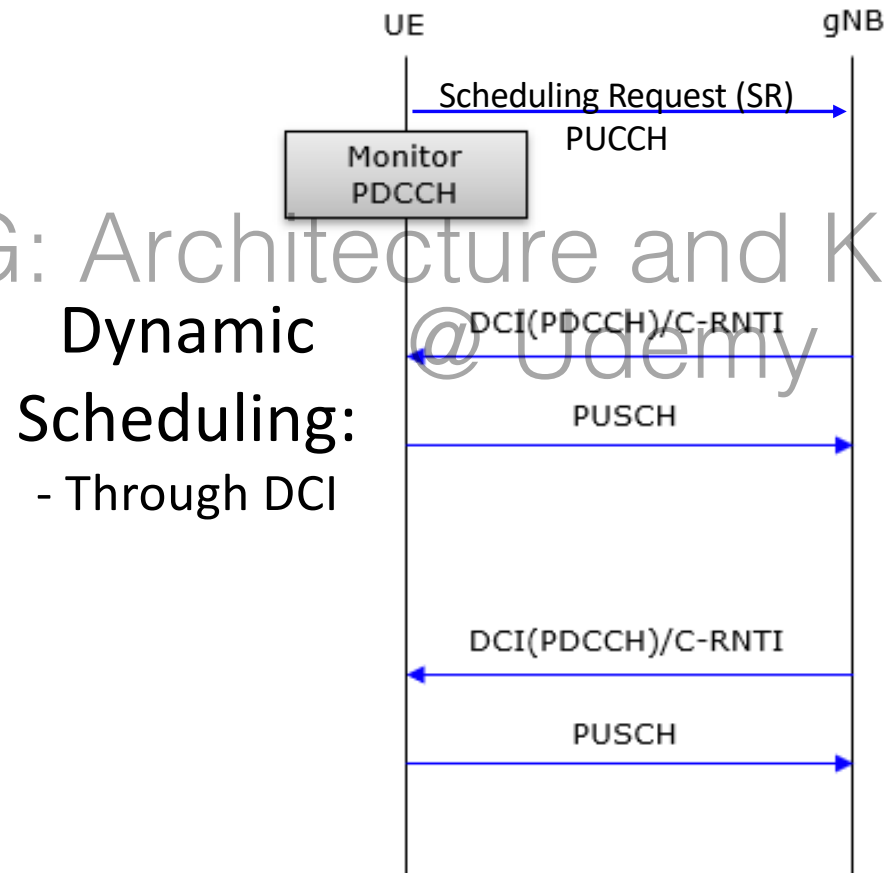


Configured Scheduling:

- Semi persistent Scheduling
- Through RRC



Uplink Scheduling



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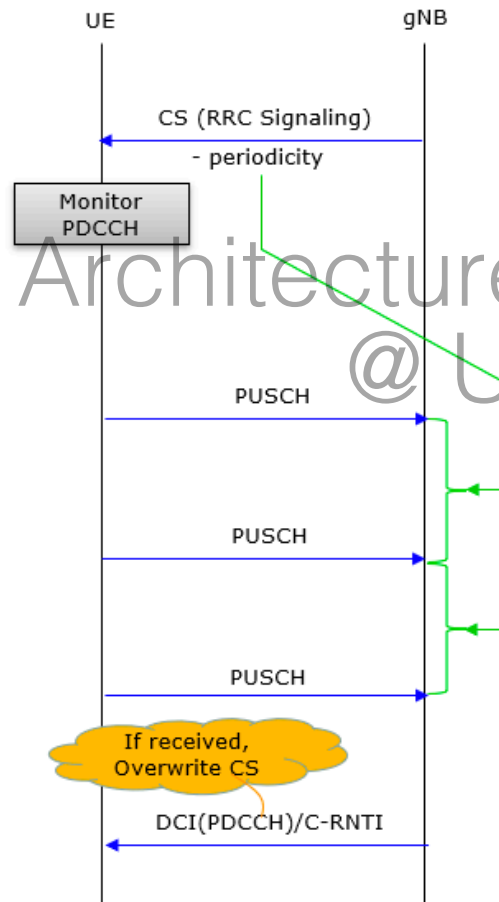
Dynamic
Scheduling:
- Through DCI



Uplink Scheduling

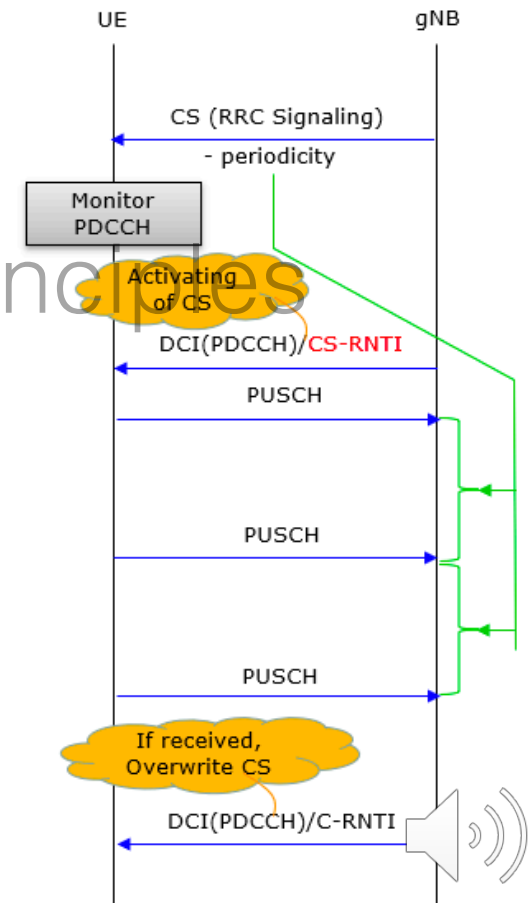
Configured Scheduling:

- Type 1: Periodic data



Configured Scheduling:

- Type 2: Activated and deactivated by DCI



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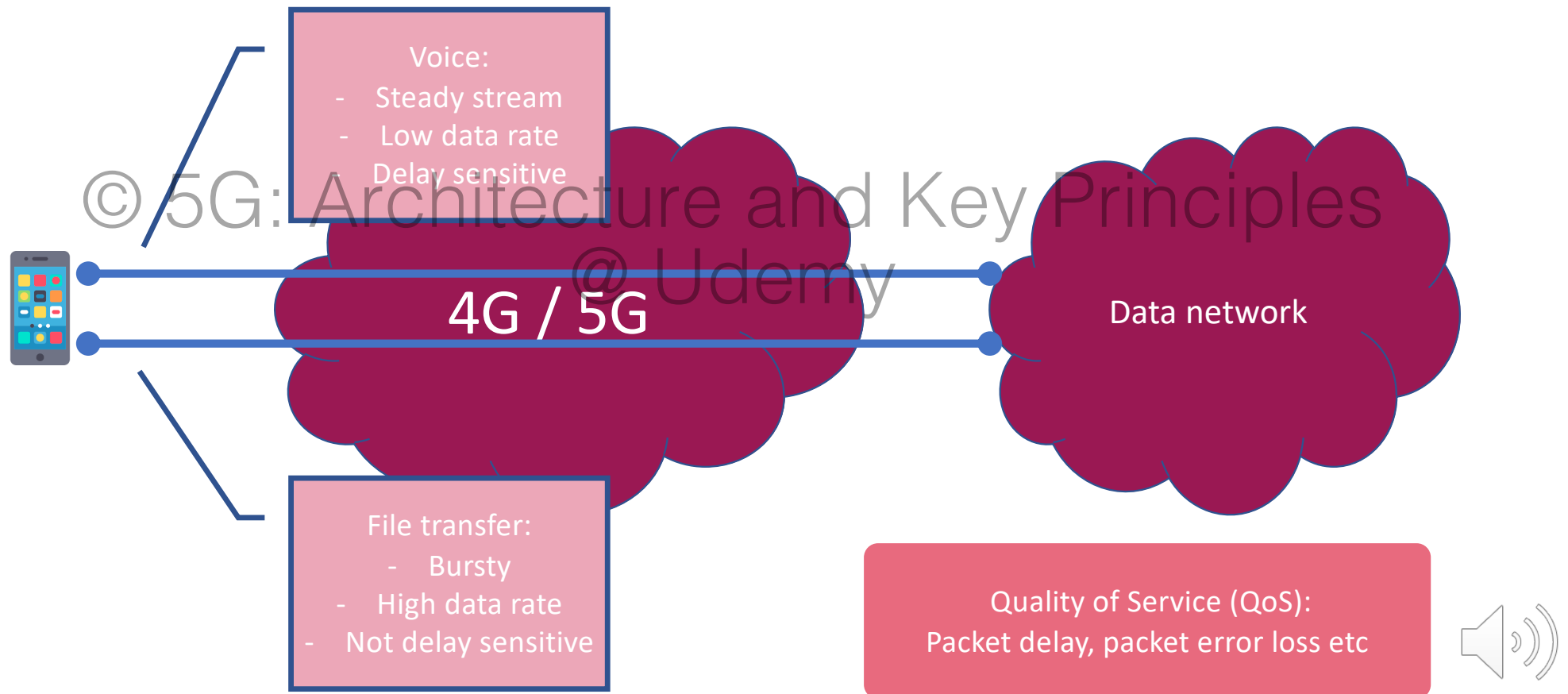


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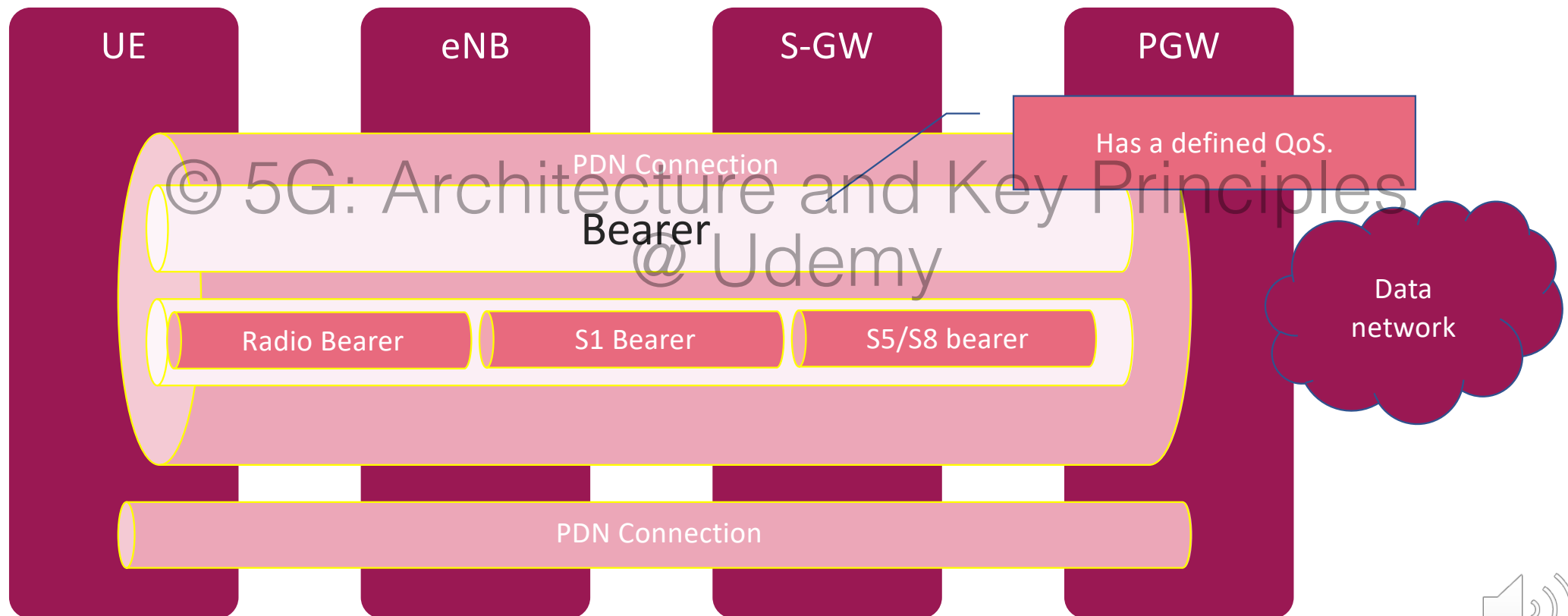
5G RAN – QUALITY OF
SERVICE (QOS)



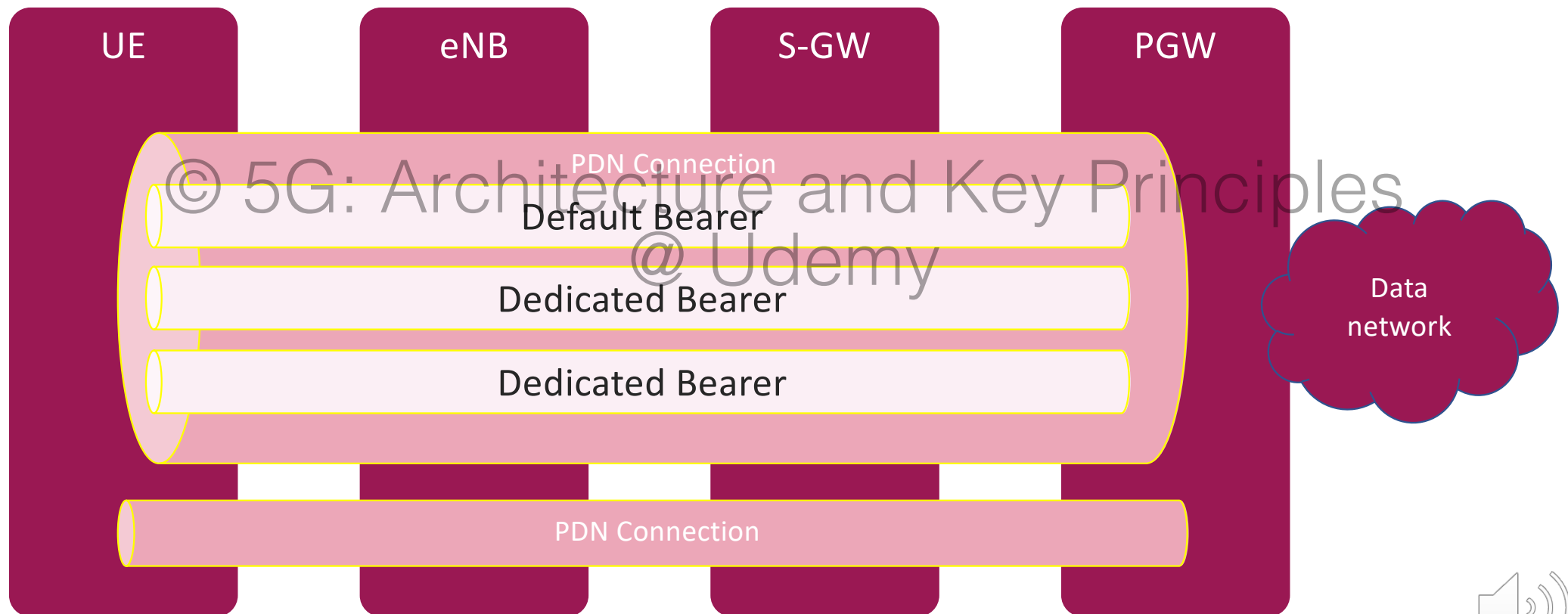
Quality of Service (QoS)



Quality of Service (QoS) in 4G



Quality of Service (QoS) in 4G

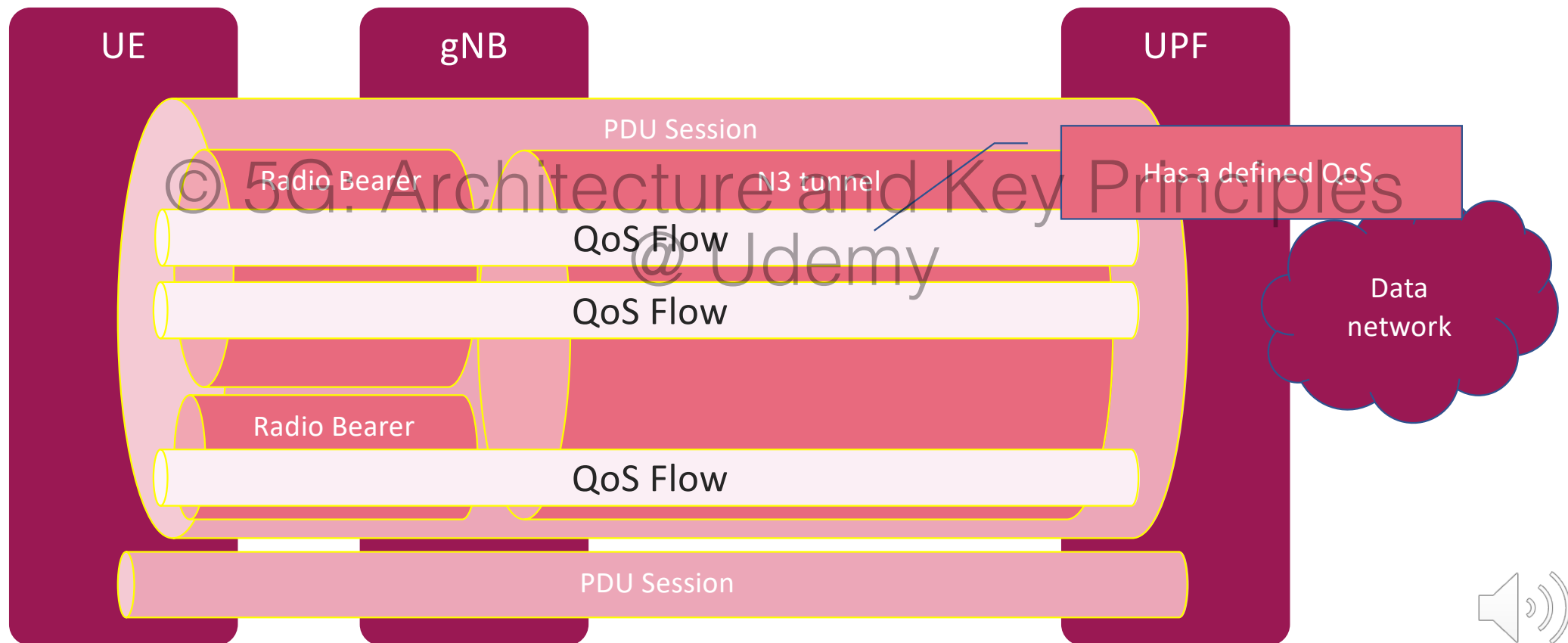


QCI in 4G

QCI	Resource Type	Priority	Packet Delay Budget (NOTE 1)	Packet Error Loss Rate (NOTE 2)	Example Services
1 (NOTE 3)	GBR	2	100 ms	10^{-4}	Conversational Voice
2 (NOTE 3)		4	150 ms	10^{-3}	Conversational Video (Live Streaming)
3 (NOTE 3)		3	50 ms	10^{-3}	Real Time Gaming
4 (NOTE 3)		5	300 ms	10^{-6}	Non-Conversational Video (Buffered Streaming)
5 (NOTE 3)	Non-GBR	1	100 ms	10^{-6}	IMS Signalling
6 (NOTE 4)		6	300 ms	10^{-6}	Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.)
7 (NOTE 3)		7	100 ms	10^{-3}	Voice, Video (Live Streaming) Interactive Gaming
8 (NOTE 5)		8	300 ms	10^{-6}	Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.)
9 (NOTE 6)		9			



Quality of Service (QoS) in 5G



5G QoS identifier (5QI) in 5G

Non-GBR (Non Guaranteed bit rate)

GBR (Guaranteed bit rate)

Delay critical GBR

5QI Value	Resource Type	Default Priority Level	Packet Delay Budget	Packet Error Rate	Default Maximum Data Burst Volume (NOTE 2)	Default Averaging Window	Example Services
1	GBR NOTE 1	20	100 ms	10^{-2}	N/A	2000 ms	Conversational Voice
2		40	150 ms	10^{-3}	N/A	2000 ms	Conversational Video (Live Streaming)
3		30	50 ms	10^{-3}	N/A	2000 ms	Real Time Gaming, V2X messages Electricity distribution - medium voltage, Process automation - monitoring
4		50	300 ms	10^{-6}	N/A	2000 ms	Non-Conversational Video (Buffered Streaming)
65		7	75 ms	10^{-2}	N/A	2000 ms	Mission Critical user plane Push To Talk voice (e.g., MCPTT)
66		20	100 ms	10^{-2}	N/A	2000 ms	Non-Mission-Critical user plane Push To Talk voice
67		15	100 ms	10^{-3}	N/A	2000 ms	Mission Critical Video user plane
75	Non-GBR NOTE 1	25	50 ms	10^{-2}	N/A	2000 ms	V2X messages
6		10	100 ms	10^{-6}	N/A	N/A	IMS Signalling
6		60	300 ms	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.)
7		70	100 ms	10^{-3}	N/A	N/A	Voice, Video (Live Streaming) Interactive Gaming
8		80	300 ms	10^{-6}	N/A	N/A	Video (Buffered Streaming) TCP-based (e.g., www, e-mail, chat, ftp, p2p file sharing, progressive video, etc.)
9		90	60 ms	10^{-6}	N/A	N/A	Mission Critical delay sensitive signalling (e.g., MC-PTT signalling)
69		5	60 ms	10^{-6}	N/A	N/A	Mission Critical Data (e.g. example services are the same as QCI 6/8/9)
70	Delay Critical GBR	55	200 ms	10^{-6}	N/A	N/A	V2X messages
79		65	50 ms	10^{-2}	N/A	N/A	Low Latency eMBB applications Augmented Reality
80		68	10 ms	10^{-6}	N/A	N/A	Remote control (see TS 22.261 [2])
81	Delay Critical GBR	11	5 ms	10^{-5}	160 B	2000 ms	



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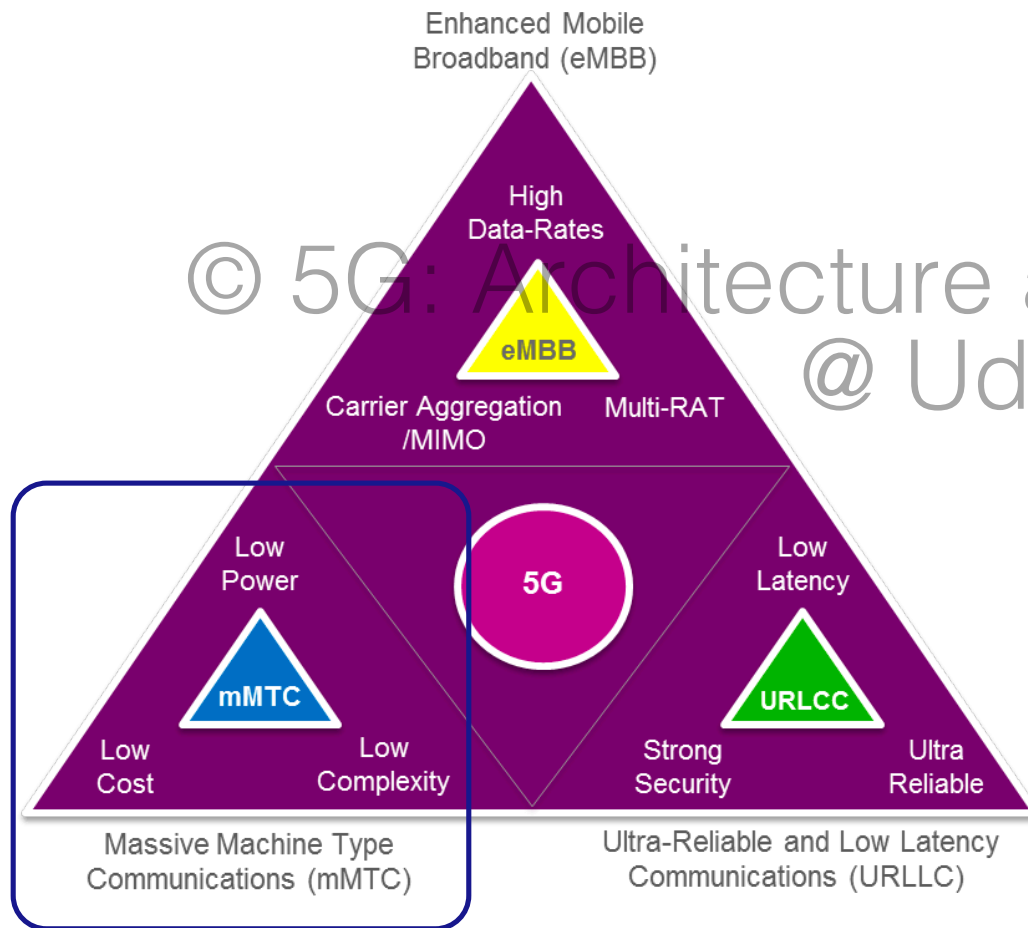


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5G RAN – MASSIVE
MACHINE TYPE
COMMUNICATIONS (MMTC)



MMTC



- Large device density (up to 50,000 devices per cell),
- Low cost
- Communicate very little data
- Long battery life (upto 10. years on 2 AA batteries) and
- Require good coverage (up to building basements).



MMTC

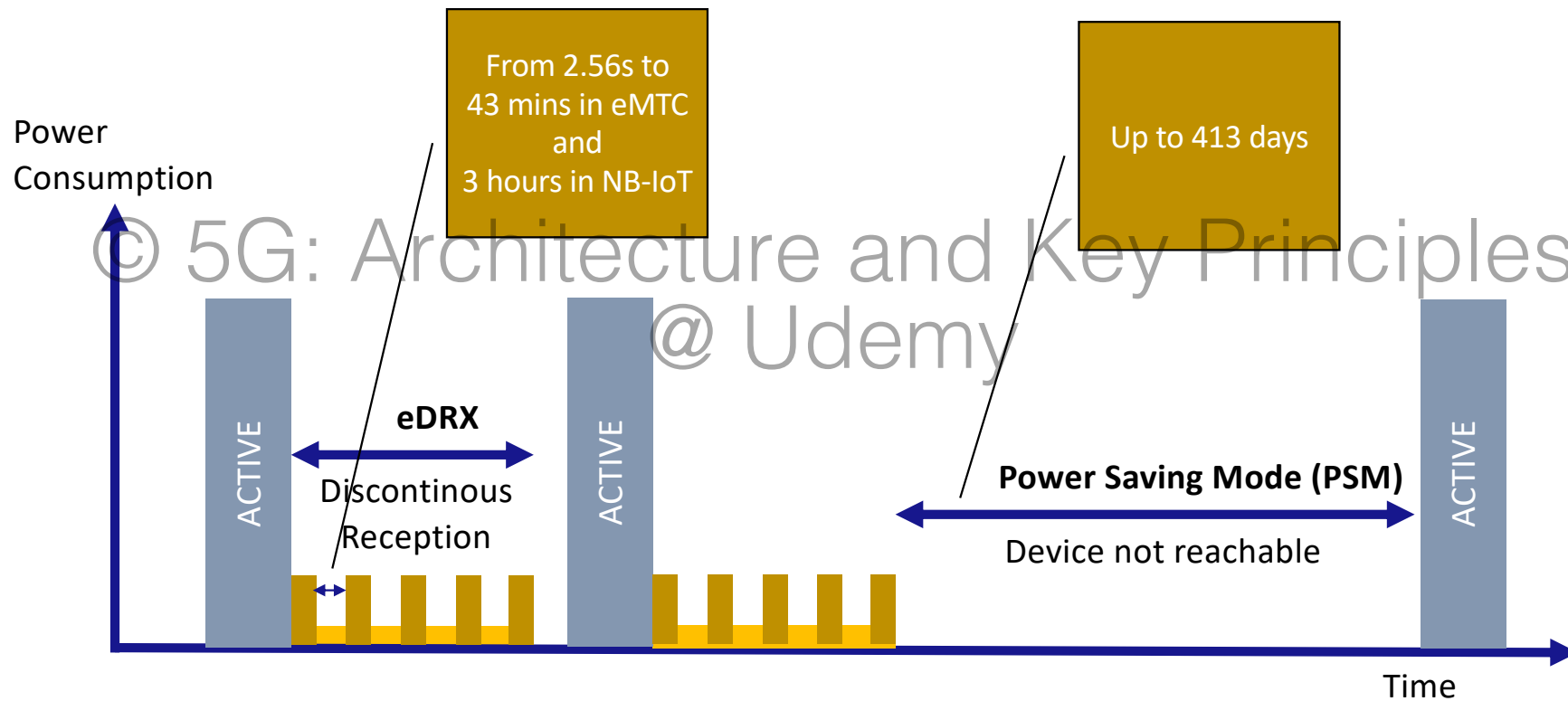


MMTC – Achieving low cost and low complexity

	Release 8 CAT-4
Downlink Peak rate	150 Mbps
Uplink Peak rate	50 Mbps
Highest DL Modulation Scheme	64 QAM
Highest UL Modulation Scheme	16 QAM
Number of receive antennas	2
UE Receive Bandwidth	20 Mhz
UE Transmit Power	23 dBm
Duplex mode	Full Duplex



MMTC – Achieving long battery life



eDRX and PSM are two key features to help achieve long battery life

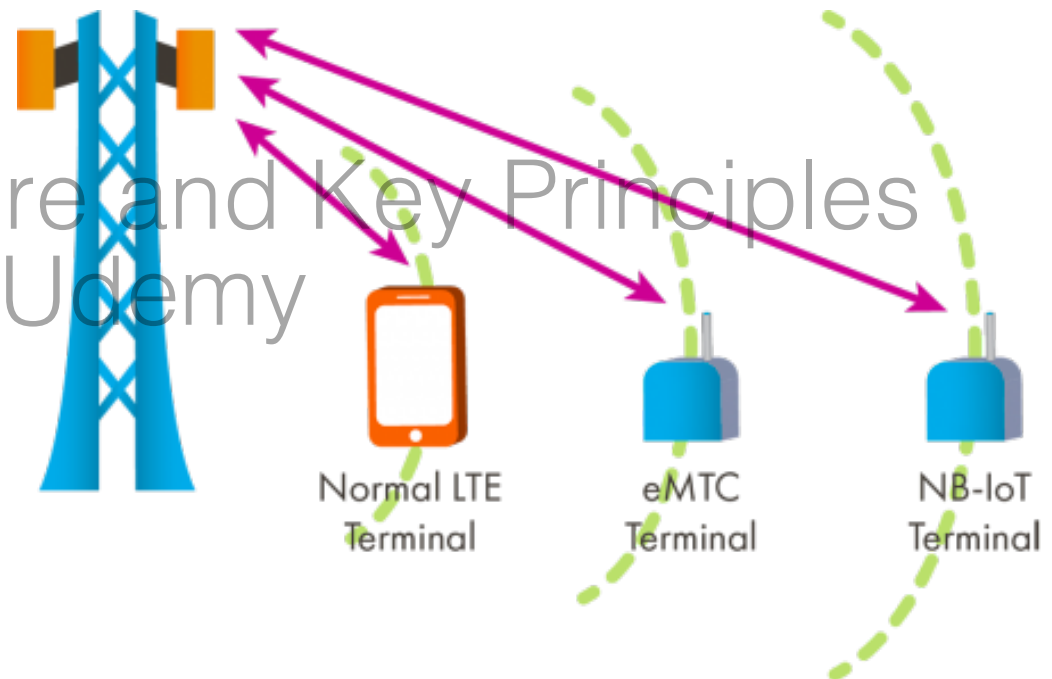


MMTC – Achieving extended coverage

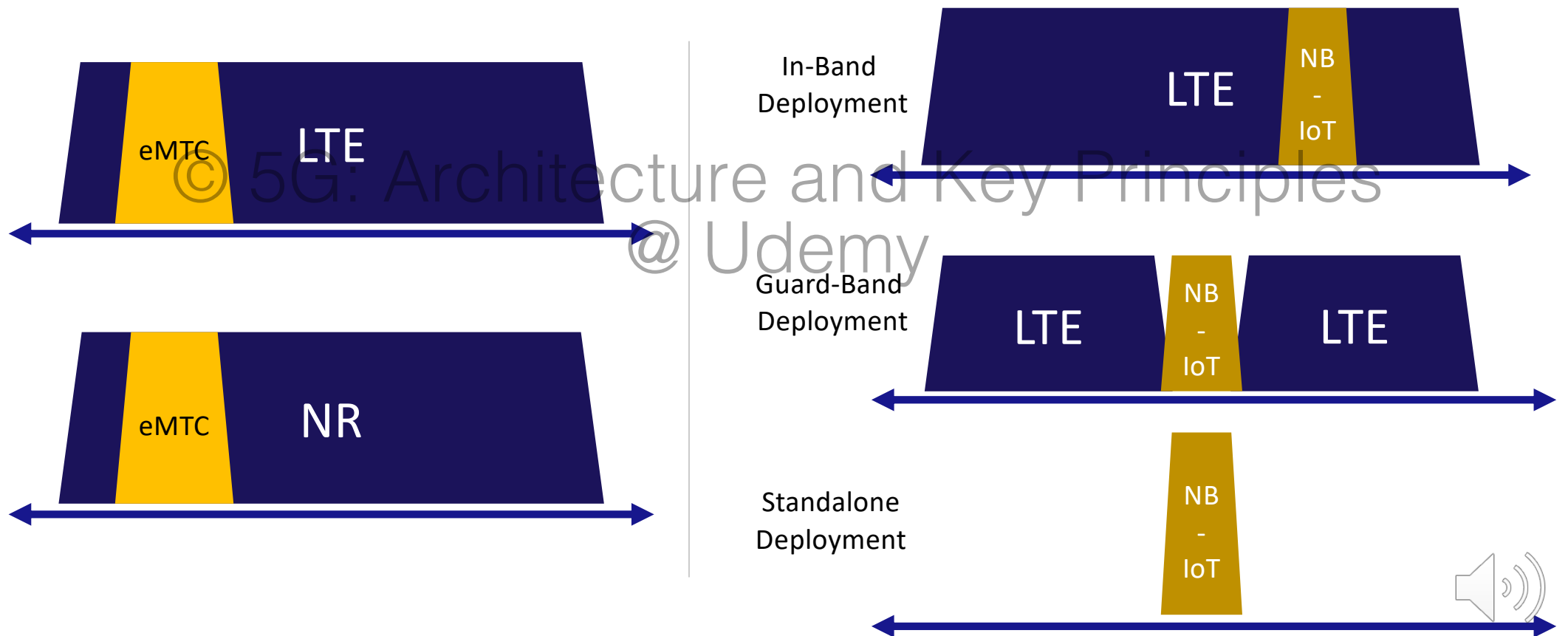
- Repetitions

- Up to 2048 repetitions in DL and 128 in UL
- eMTC: 10 dB over GPRS, 15 dB over LTE
- NB-IoT: 20 dB over GPRS, 25 dB over LTE

- Higher power in control channels



Deployment of MMTC Technologies



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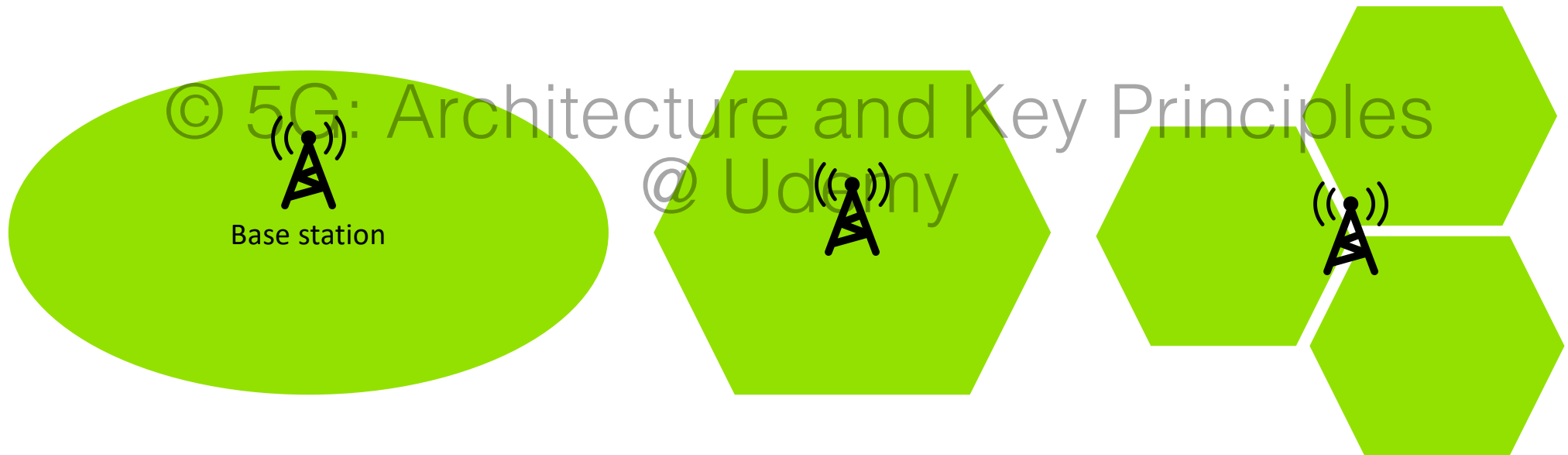


5G:
Architecture
and Key
Principles

5G RAN – NR CELL



Concept of cell



Concept of cell – Cell Search

Cell search:

- Synchronization
- System information



Concept of cell – Cell Search

LTE Cell

- Primary Synchronization Signal (PSS)
- Secondary Synchronization signal (SSS)
- Cell System Information
 - Master Information Block (MIB)
 - System Information Block (SIB)
 - SIB1
 - SIB2
 -
 - SIB9

NR Cell

- Primary Synchronization Signal (PSS)
- Secondary Synchronization signal (SSS)
- Cell System Information
 - Master Information Block (MIB)
 - System Information Block (SIB)
 - SIB1
 - SIB2
 -
 - SIB9

Cell specific
information

Concept of cell

LTE Cell

- Primary Synchronization Signal (PSS)
- Secondary Synchronization signal (SSS)
- Cell System Information
 - Master Information Block (MIB)
 - System Information Block (SIB)
 - SIB1
 - SIB2
 -
 - SIB9

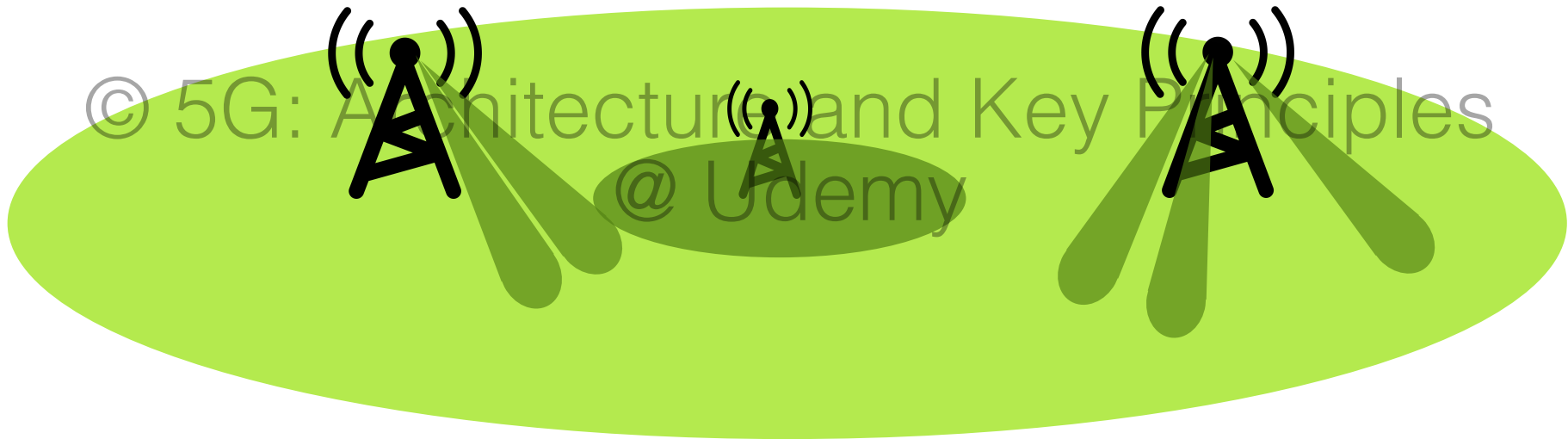
NR Cell

- Primary Synchronization Signal (PSS)
- Secondary Synchronization signal (SSS)
- Cell System Information
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 - SIB1
 - SIB2
 -
 - SIB9

SS Block
(SSB)



Concept of cell



Different tx/rx points, same SSB => same NR cell



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